

Expanding Agricultural Exports of Arunachal Pradesh Through Infrastructure Development



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List of Abbreviations

Abbreviation	Full Form
APAMB	Arunachal Pradesh Agriculture Marketing Board
APEDA	Agricultural and Processed Food Products Export Development Authority
CA	Controlled Atmosphere
FPO	Farmer Producer Organization
FYM	Farm Yard Manure
GI	Geographical Indication
Gol	Government of India
ICAR	Indian Council of Agricultural Research
IQF	Individual Quick Freezing
KVK	Krishi Vigyan Kendra
MIDH	Mission for Integrated Development of Horticulture
MOVCD-NER	Mission Organic Value Chain Development for North Eastern Region
NABARD	National Bank for Agriculture and Rural Development
NEDFi	North Eastern Development Finance Corporation Ltd.
NER	North Eastern Region
NPOP	National Programme for Organic Production
PGS-India	Participatory Guarantee System – India
PKVY	Paramparagat Krishi Vikas Yojana
RKVY	Rashtriya Krishi Vikas Yojana
WRS	Warehouse Receipt System

Contents

Part 1. The Context	9
1. Introduction	9
2. Arunachal Pradesh state profile	10
3. Agriculture in Arunachal Pradesh	11
4. Horticulture in Arunachal Pradesh	14
Summary	15
Part 2. Important Agriculture Products of Arunachal Pradesh	16
5.1 Non-Perishable Produce	16
5.2 Item wise description of Non-Perishable Produce: Arunachal Pradesh in comparison with NER	18
5.2.1 Paddy	18
5.2.2 Maize	18
5.2.3 Pulses	19
5.2.4 Oilseeds	19
5.3 Perishable Produce	20
5.4 Item wise description of Perishable Produce: Arunachal Pradesh in comparison with NER	21
5.4.1 Kiwi	21
5.4.2 Orange	22
5.4.3 Pineapple	22
5.4.4 Apple	23
5.4.5. Potato	23
5.4.6. Large Cardamom	23
5.4.7. Ginger	24
Summary	24
5.5 Work Plan for the identified Agriculture and Horticulture Produce of Arunachal Pradesh	26
Summary	27
6.1. Organic Agriculture in Arunachal Pradesh: A Strategic Opportunity	27
6.1.1 Organic Certification and other Government Initiatives	27
6.1.2 Status of Organic Farming in Arunachal Pradesh	28
Part 3. Infrastructure Requirements for Agricultural Produce	30
7.1 Infrastructure Gap Assessment	30
7.1.1 Critical Infrastructure gaps affecting perishable produce	31
7.1.2 Infrastructure gaps affecting non-perishable produce	32
7.2 Differentiating Infrastructure Needs for Perishable and Non-Perishable Produce	32

PART 4 Recommendations	35
8.1 Recommendations for Agricultural Infrastructure Development in Arunachal Pradesh ..	35
8.1.1 Articulate clear goals and objectives	35
8.1.2 Undertake strategic planning of investments and institutional support	36
8.1.3 Sustain institutional support to farmers	38
8.1.4 Improve market access and facilitation	39
Annexure 1	40
Kiwi: Supply chain and Logistics Infrastructure	40
Annexure 2	46
Case examples of successes and challenges in yak churpi, kiwi wine, organic and mushroom farming in Arunachal Pradesh.....	46
Annexure 3	49
Primary Processing and Storage Facilities	49
Annexure 4	53
Transport Connectivity	53
Annexure 5	55
State-wise description of the main fruits, vegetables and spices produced in the NER.....	55
Annexure 6	56
State-wise distribution of numbers of operational holdings and area operated in the NER – all social groups	56
Annexure 7	57
State-wise Area and Production of Horticulture Crops from the year 2011-12 to 2020-21 ..	57

List of Figures

Figure 1. Administrative Map of Arunachal Pradesh	10
Figure 2. Land Use Pattern of Arunachal Pradesh.....	11
Figure 3. Agriculture Land Use in Arunachal Pradesh (2023).....	12
Figure 4. Farm Holding Pattern in Arunachal Pradesh.....	12
Figure 5. Chemical Fertilizer Use In Arunachal Pradesh	13
Figure 6. Share of Area under Horticulture of Arunachal Pradesh vs India	15
Figure 7. Horticulture Productivity across States and National Average.....	15
Figure 8. Non- Perishable Produce in Arunachal Pradesh	17
Figure 9. Top Non-Perishable Produce producing districts in Arunachal Pradesh.....	17
Figure 10. Top Perishable Produce producing districts in Arunachal Pradesh	20
Figure 11. Import of Kiwi in India and production trends in Arunachal Pradesh	41
Figure 12. Kiwi productivity comparison among kiwi producing states in India	41
Figure 13. Kiwi producing states in India. The darker the colour higher the production	42
Figure 14. Transport route for kiwi from Ziro valley to Guwahati	44
Figure 15. Map of infrastructure gaps in Kiwi supply chain	45

List of Tables

Table 1. Paddy Production and Productivity	18
Table 2. Maize Production and Productivity.....	18
Table 3. Pulses Production and Productivity	19
Table 4. Oilseeds Production and Productivity	20
Table 5. Kiwi Production and Productivity	21
Table 6. Orange (including the GI tagged Wakro Orange) Production and Productivity	22
Table 7. Pineapple Production and Productivity	22
Table 8. Apple Production and Productivity.....	24
Table 9. Potato Production and Productivity	24
Table 10. Large Cardamom Production and Productivity	24
Table 11. Ginger Production and Productivity	24
Table 12. The crop-wise potential of area expansion of the focus crops in the state (NEDFi (2017)	25
Table 13. Work Plan for Non-Perishable Crop Clusters in Arunachal Pradesh.....	26
Table 14. Work Plan for Perishable Crop Clusters in Arunachal Pradesh	26
Table 15. Cold Storage Infrastructure in Northeast India	31
Table 16. Differentiating Infrastructure Needs for Perishable and Non-Perishable Produce	32
Table 17. Opportunities for investment in Perishable and Non-Perishable produce in Arunachal Pradesh	49
Table 18. Agri processing/Industrial Cooperative in Arunachal Pradesh	50
Table 19. Proposed aggregation centres at identified clusters (NEDFi)	51
Table 20. Major routes and the districts for Agriculture produce	53
Table 21. Key agricultural regions and travel estimates.....	53



Photo: Wet rice cultivation in Arunachal Pradesh

Part 1. The Context

1. Introduction

Geopolitical Significance: The Northeastern Region (NER) of India comprises eight states—Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, and Tripura. The NER covers approximately 262,184 square kilometres, accounting for about 8% of India’s total area. It shares an extensive international boundary of approximately 5,812 kilometres with five countries: China, Myanmar, Bangladesh, Bhutan, and Nepal¹. Its long, porous borders and proximity to several countries' markets could be advantageous for development of an export hub. Strategically located, the region holds significant importance in India’s Act East Policy and its engagement with ASEAN countries. It is a vital yet underutilized region in terms of both strategic location and agricultural development.

Natural Advantages and Organic Produce: The region possesses several natural and comparative advantages. Its fertile soils, rich in organic matter, coupled with a favourable climate, support the cultivation of a wide range of exotic crops. It is also one of India’s richest biodiversity zones. Although nearly two-thirds of the terrain is hilly—limiting the potential for large-scale industrialization and conventional agriculture—the diversity in soil and agro-climatic conditions give the NER a distinct edge in agriculture production. The region supports the cultivation of tropical, sub-tropical, and temperate fruits and vegetables, many of which are grown with minimal or no chemical inputs. As a result, much of NER’s produce is organic by default.

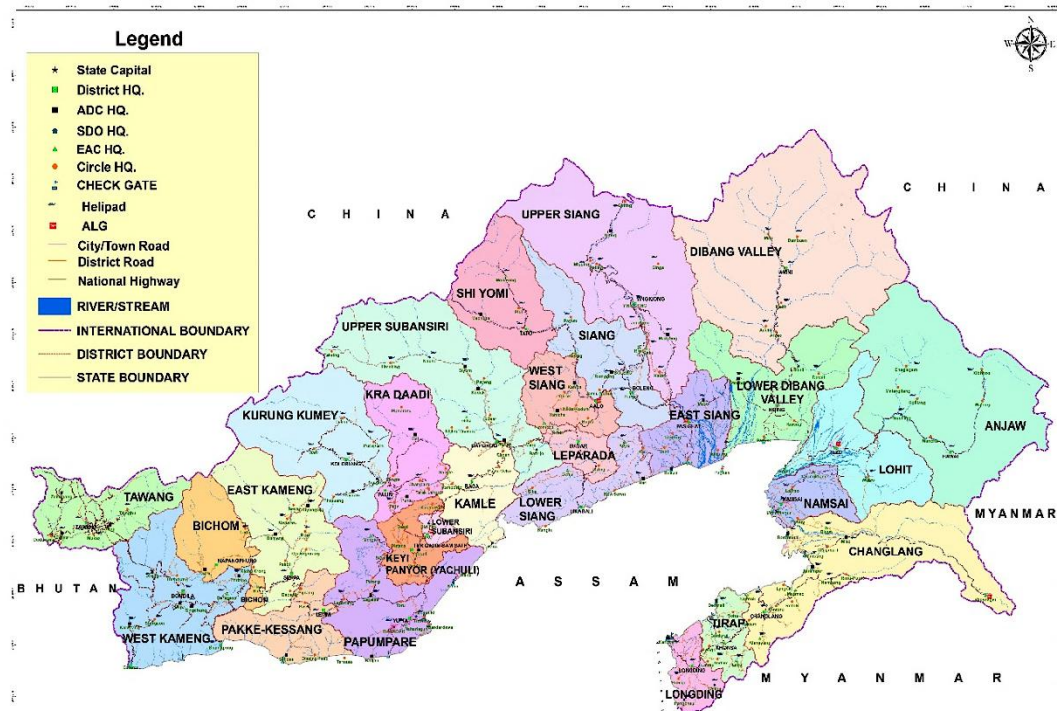
¹ <https://www.mha.gov.in/en/commoncontent/north-east-division>

With increasing income levels and rising health consciousness, the demand for organic produce is growing both in domestic and international markets. This presents a unique opportunity to promote the organically grown products of the region and integrate them into wider value chains.

2. Arunachal Pradesh state profile

Arunachal Pradesh shares a long international border with China (1,080 km) to the north and northeast, Bhutan (160 km) to the west, and Myanmar (440 km) to the east. The state stretches from the Brahmaputra valley plains in the south to the snow-capped peaks of the eastern Himalayas. It has the largest area among the Seven Sister States of the Northeast. Arunachal Pradesh spans an area of 83,743 sq.km., accounting for approximately 2.55% of India's total land area, however 93% of the area is under forest. It is traversed by five major rivers — the Kameng, Subansiri, Siang, Tirap, and Lohit — along with countless smaller rivers and streams, many of which are fed by snowmelt. There are 27 districts which was earlier 16 districts as per the Census 2011, the latest 27th district Bichom, carved out of West and East Kameng in March 2024² (**Figure 1**). The state has the highest hydropower availability in the NER and aims to develop 30,000 MW of hydroelectric capacity by 2030.

Figure 1. Administrative Map of Arunachal Pradesh



Arunachal Pradesh's regions are separated by vast differences in altitude and topography, from the low-lying foothills of Itanagar (320 meters above sea level) to the high mountains of Tawang (3,048 meters above sea level) and the remote valleys of Vijaynagar, 1,240 meters above sea level and bordered by Myanmar on three sides. The state's connectivity is hindered by its mountainous, forested terrain, making travel time-consuming and often requiring overnight stops, e.g. the journey to Tawang from Guwahati typically requires a night halt at Dirang.

The 2011 Census recorded a population of 1,383,727 (13.84 lakh). The projected population of Arunachal Pradesh in 2025 is estimated to be between 1.59 million and 1.89 million (15.89

² The Hindu (8th March 2024, available at <https://www.thehindubusinessline.com/news/national/bichom-becomes-27th-district-of-arunachal-pradesh/article67927995.ece>)

lakh to 18.91 lakh³. The average population density is about 17 persons per square km, which is sparse compared to the national average 382 people per square km.

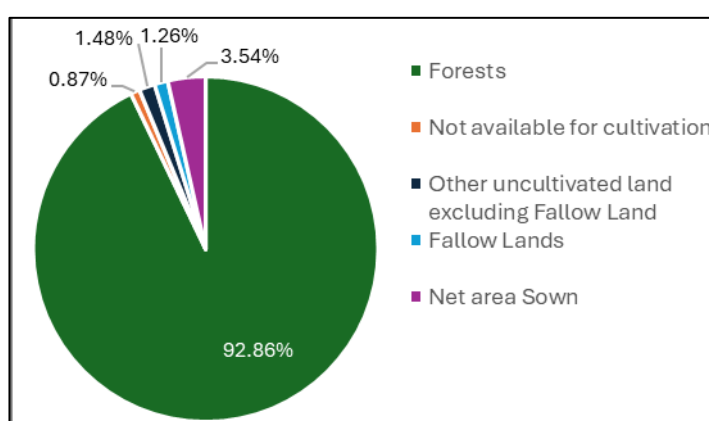
3. Agriculture in Arunachal Pradesh

Agriculture forms the backbone of Arunachal Pradesh's economy. Although around 60% of its population rely on agriculture for livelihood and food security, its contribution to the Gross State Domestic Product (GSDP) has declined—from approximately 40% in 1990-91 to 30% in 2020-21. This decline points to low agricultural productivity and poor income levels among the population solely engaged in agriculture.

The services sector has seen a notable rise in its share of the GSDP. Between 2016-17 and 2020-21, its share has increased from 38.8% to 40.66% (State Finances Audit Report, 2022)⁴. By 2021-22, the services sector contributed 40.2% to Arunachal Pradesh's Gross State Value Added (GSVA), making it the largest sector in the state's economy. Key drivers within services include public administration, other services, and trade, hotels, and restaurants.

Figure 2 illustrates the land use pattern in the state. Only 3.54% is utilized as net sown area. This is because of hilly and highly forested terrain and low population density. Given the limited availability of agricultural land, it is essential to ensure efficient utilisation by strengthening infrastructure and agricultural practices.

Figure 2. Land Use Pattern of Arunachal Pradesh



Source: APEDA, Ministry of Agriculture & Farmers Welfare 2023-24

Arunachal Pradesh has three agro-climatic zones — Temperate, Sub-tropical and Tropical — that can support cultivation of a multitude of crops, vegetables and fruits. The diverse soil types, including red-loamy and laterite, support a variety of crops. Major crops include paddy (including sticky rice), millet, wheat, pulses, potatoes, sugarcane, oilseeds, and maize. The state's horticultural crops include citrus, kiwi, spices, and medicinal plants, are a resource with significant economic potential.

Arunachal Pradesh is the largest producer of kiwi, second-largest producer of large (black) cardamom, in India⁵. The state is known for its high-quality citrus fruits and has received a

³ <https://www.census2011.co.in/census/state/districtlist/arunachal+pradesh.html>

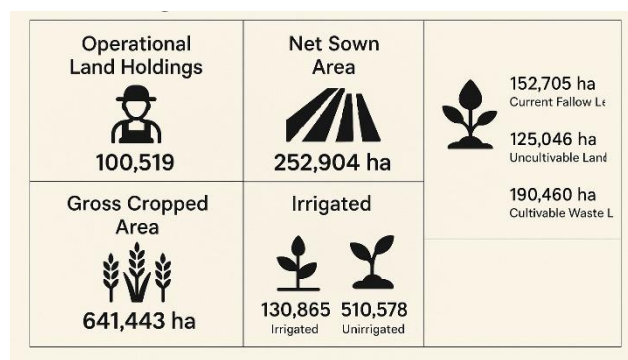
⁴ [https://cag.gov.in/webroot/uploads/download_audit_report/2023/SFR-State-Finance-Report-2021-22-\(14-02-2023\)-FINAL-FOR-PRINT-0642534167811c7.85295835.pdf](https://cag.gov.in/webroot/uploads/download_audit_report/2023/SFR-State-Finance-Report-2021-22-(14-02-2023)-FINAL-FOR-PRINT-0642534167811c7.85295835.pdf)

⁵ Ministry of Development of Northeastern Region, Government of India (available at <https://risingnortheast.in/arunachal-pradesh/#:~:text=Did%20you%20know?,region%20in%20terms%20of%20area.&text=It%20is%20also%20the%20n>)

GI tag for the Arunachal Mandarin Orange (Wakro Orange), renowned for its bright colour and quality. In addition, Arunachal Pradesh also has GI tags for Khaw Tai (Khamti local rice), Adi Kekir (Ginger), and Yak Churpi (naturally fermented milk product made from yak milk)

Tropical fruits like pineapples, bananas, and oranges are cultivated in the valleys, while apples, peaches, and plums are grown in the colder, high-altitude areas. Non-traditional crops such as tea, cashew nuts, oilseeds, tomatoes, mushrooms, and wheat have also been grown successfully in the state.

Figure 3. Agriculture Land Use in Arunachal Pradesh (2023)



The agricultural land use data highlights 1,13,253 operational land holdings covering a net sown area of 2,52,904 hectares. **(Figure 3)** The gross cropped area stands significantly higher at 6,41,443 hectares, indicating widespread multiple cropping. Irrigated land accounts for 1,30,865 hectares, while 5,10,578 hectares remain unirrigated, showing a major dependence on rainfall. 1,52,705 hectares of land is lying fallow, and 190,460 hectares is cultivable waste land, which signifies considerable potential for expansion of agricultural activity through improved land management.

Figure 4. Farm Holding Pattern in Arunachal Pradesh

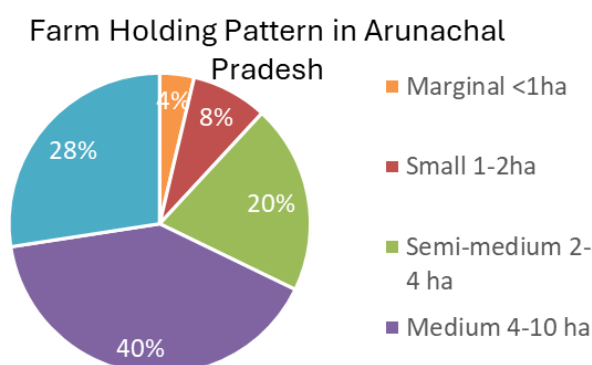
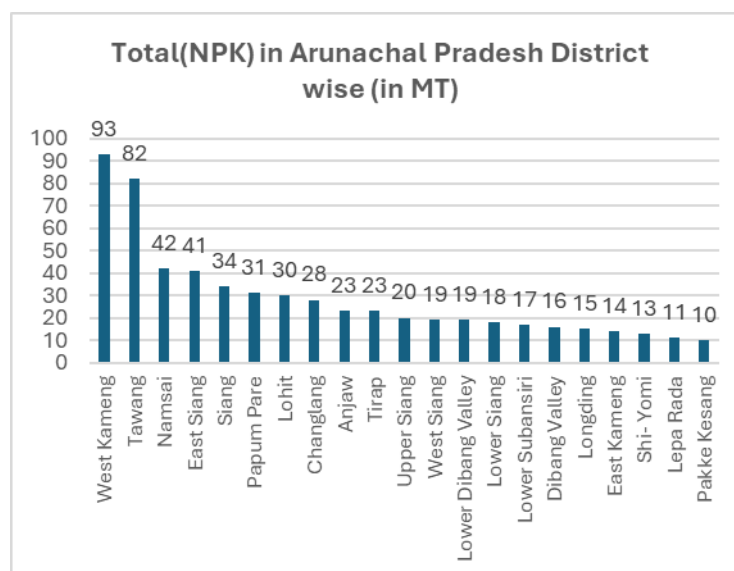


Figure 4 shows the distribution of farm holdings by size, with 60% of holdings falling in the 2–10-hectare range. In contrast, marginal (<1 ha) and small (1–2 ha) holdings together make up just 12% of the total. Most of the farm holding is medium-sized holdings.

The State Government has launched key initiatives like the ‘Atma Nirbhar Krishi Yojna’ for farmers and the ‘Atma Nirbhar Bagwani Yojna’ for horticulture, aimed at promoting self-reliance in agriculture.

Arunachal has a strategic location with extensive international borders enhancing the potential for trade with ASEAN countries. Recognized as a ‘fast mover’ in the North-Eastern Region by the DPIIT's LEADS ranking and ranked 8th out of 9 states in NITI Aayog's Export Preparedness Index 2022⁶ for Himalayan states, Arunachal Pradesh can steadily build its trade and export capacity.

Figure 5. Chemical Fertilisers Use in Arunachal Pradesh



Source: Directorate of Agriculture, Government of Agriculture 2022-23

The use of chemical fertilizers has a direct impact on export potential. **Figure 5** represents the district-wise usage of chemical fertilizers (Nitrogen, Phosphorus, and Potassium). There is a stark disparity across districts, with the top two districts West Kameng and Tawang using about double the amount used in the third and fourth highest, Namsai and East Siang.

The overall consumption of chemical fertilizers in Arunachal Pradesh is extremely low when compared to other states and the national average. The state reported a total NPK consumption of 599 MT in 2023–24, compared to India's total consumption of 30.64 million MT. The average consumption across Indian states is around 137.15 kg/ha, while Arunachal Pradesh's average is 2.115 kg/ha (2021)⁷.

Given its extremely low usage of NPK fertilizers Arunachal Pradesh is organically inclined by default. The state's organic potential is further supported by its rich biodiversity and traditional farming systems that inherently rely on natural inputs. This presents a unique advantage for the state in aligning with global organic and sustainable farming trends, as minimal chemical input is a prerequisite for attaining certified organic status.

The Government of India has been encouraging organic farming through various schemes for capitalizing the benefits of organic farming, enhancing soil health and ensuring the

⁶ NITI Aayog (2022), Export Preparedness Index 2022, available at https://www.niti.gov.in/sites/default/files/2023-07/Export-Preparedness-Index-2022_0.pdf

⁷ Directorate of Economics and Statistics, Department of Agriculture and Farmers Welfare, Available at <https://www.ceicdata.com/en/india/chemical-fertilizers-nitrogen-phosphate-and-potash-npk-consumption-per-hectare-by-states/chemical-fertilizers-npk-consumption-per-hectare-arunachal-pradesh#:~:text=ha%20for%202020,-.Chemical%20Fertilizers:%20NPK%20Consumption:%20Per%20Hectare:%20Arunachal%20Pradesh%20data,0.000%20kg/ha%20in%202021.>

production of pesticide-free food. Arunachal Pradesh can avail benefits under major schemes such as Mission Organic Value Chain Development Northeastern Region (MOVCDNER), Parampragat Krishi Vikas Yojana (PKVY), Rashtriya Krishi Vikas Yojana (RKVY), Mission for Integrated Development of Horticulture (MIDH).

Government of India support can help the state overcome some of its key challenges, such as low agricultural productivity due to limited mechanization, shortage of high yielding and climate-resilient seeds, and lack of transportation and marketing infrastructure. Factors such as inconsistent product quality standards, inadequate market intelligence and linkages, and supply chain disruptions due to production volatility continue to constrain the state's export potential.

4. Horticulture in Arunachal Pradesh

The cultivation of horticultural crops is important for Arunachal Pradesh, however according to a study conducted by NEDFi⁸, the current production levels of various crops in the region are insufficient to achieve economies of scale. Only pineapples, oranges and apples are produced in substantial quantities, whereas production of other fruits like kiwi, passion fruit and walnut needs considerable enhancement. The production of vegetables like potato and tomato and spices like ginger and large cardamom can be significantly enhanced through investment in processing. However, given the potential for higher value-added production in horticulture, it is a matter of concern that the area under horticulture cultivation in Arunachal Pradesh has declined between 2010-11 and 2020-21, although it has expanded at the national level.

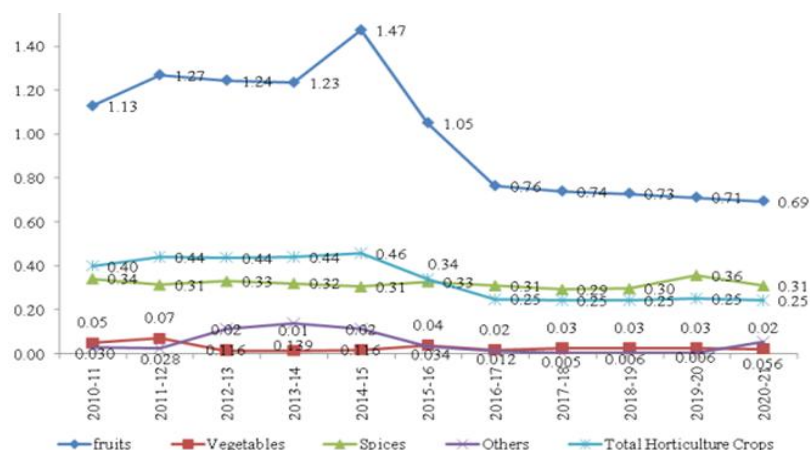
A study by Melo et.al⁹ analysed that between 2010–11 and 2020–21, Arunachal Pradesh's overall share in the country's horticultural crop area declined from 0.40% to 0.25%, with notable decrease in its share of fruits, vegetables, and spices. However, the state recorded a marginal increase in its share of area under 'other crops,' indicating a shift in cropping patterns.

The profile of horticulture production in the state is illustrated through three figures on the subsequent page. **Figure 6** shows the area under cultivation and the production of various horticultural categories. Despite covering a relatively modest land area, fruits show high production. Spices occupy a smaller cultivation area but have high production and value of output, second only to fruits. The overall productivity of horticulture crops in Arunachal Pradesh is lower than the average in the Northeastern States (**Figures 7**).

⁸ Northeastern Development Finance Corporation Ltd (NEDFi) (2022), Study and Action Plan for promoting fruits & vegetables processing industries in the state of Arunachal Pradesh. Available at <https://www.nedfi.com/wp-content/uploads/2022/06/COVERES-2.pdf>

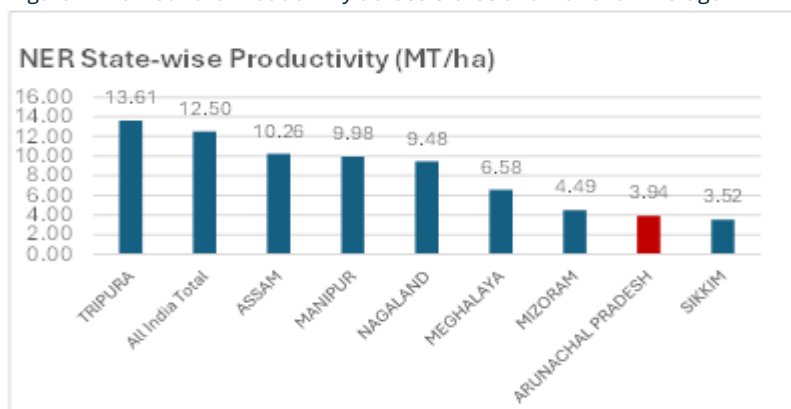
⁹ Melo, Y and Das, AK (2022), Horticulture Crops in Arunachal Pradesh: Growth, Contribution to State Economy and Decomposition Analysis. Available at <https://www.socialchangeanddevelopment.in/downloads/july2022/paper-5.pdf>

Figure 6. Share of Area under Horticulture of Arunachal Pradesh vs India



Source: Data for 2010-11 to 2012-13 are taken from Government of India (2013), for 2013-14 to 2017-18 taken from Government of India (2018) and for 2018-19 to 2020-21 taken from Ministry of Farmers Welfare Government of India (2022)

Figure 7. Horticulture Productivity across States and National Average



Source: Source: National Horticulture Board 2022-23, GOI

Summary

Arunachal Pradesh, with its vast geographic expanse, agro-climatic diversity, and strategic location, holds immense potential to become a key player in India's agricultural and horticultural economy. Despite the state's heavy reliance on agriculture for livelihoods, productivity remains low due to infrastructural gaps, limited mechanization, and underutilized land resources. The region's natural advantages—organic-rich soils, biodiversity, and suitability for a wide variety of crops—offer a strong foundation for growth, especially in horticulture. Notably, fruits and spices demonstrate high total production despite smaller cultivation areas, highlighting their economic importance. However, yields remain low, which can be potentially due to inefficiencies in technology adoption, input supply, and crop management practices that must be addressed through targeted investments in technology, improve varieties of inputs, better irrigation, post-harvest infrastructure, market linkages, and capacity building and training. With strategic interventions, Arunachal Pradesh can harness its unique strengths to become a sustainable, organic-driven agri-export hub within India's Act East framework.



Photo: A bangra pineapple (*Ananas comosus*) ready for harvest

Part 2. Important Agriculture Products of Arunachal Pradesh

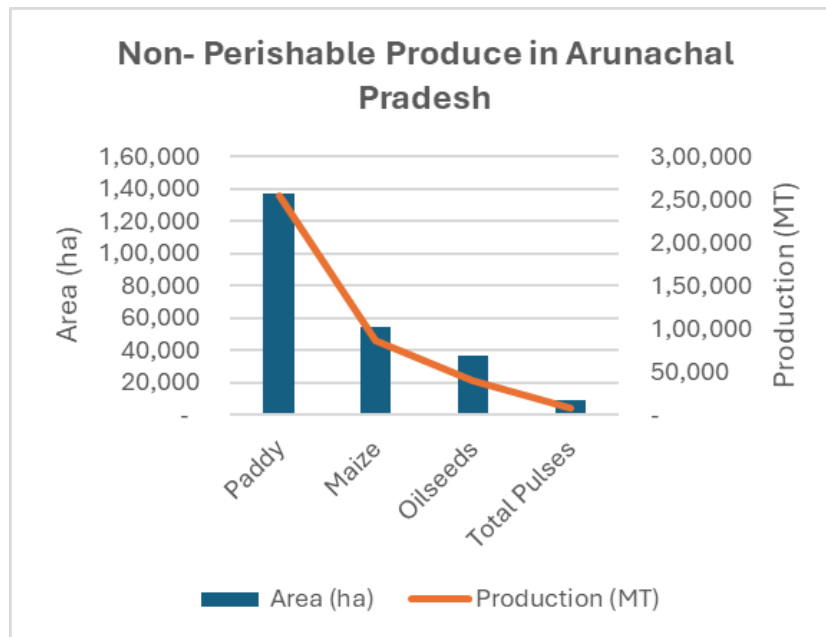
5.1 Non-Perishable Produce

Non-perishable produce consists of food items that can be stored at room temperature for long periods without spoiling. Examples include rice, beans, grains. Non-perishable foods do not require refrigeration and have a much longer shelf life compared to perishable foods.

Arunachal Pradesh supports the cultivation of a wide variety of non-perishable field crops. These include cereals, millets, pulses, oilseeds, spices, and tubers. These crops form the backbone of the state's agrarian economy and food security system. Understanding the area under cultivation and production volume of these crops is crucial for planning post-harvest infrastructure, cold chain logistics, and value addition interventions.

Figure 9 illustrates the area under cultivation and production of key perishable crops grown in the state. Paddy is an item of staple consumption, is the major crop and is the highest cultivated crop reflected both in terms of area (1.37 lakh ha) and production (2.54 lakh MT).

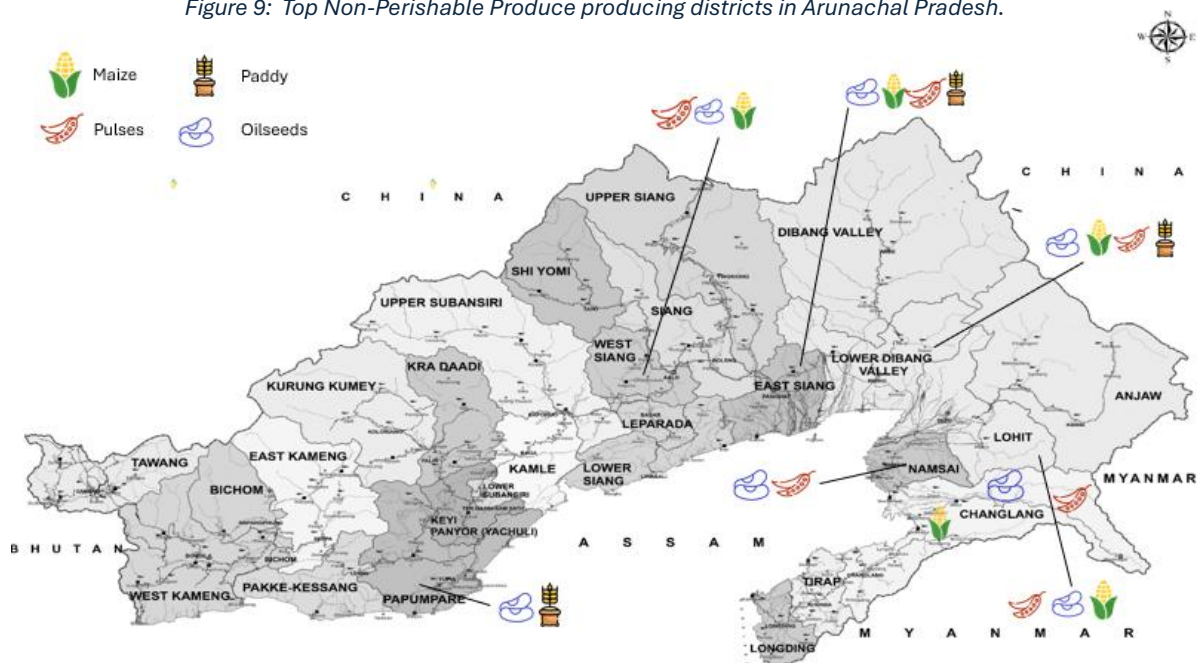
Figure 8: Non-Perishable Produce in Arunachal Pradesh



Source: Directorate of Economics and Statistics 2022-23, Ministry of Agriculture and Farmer Welfare

The second largest crop is Maize, with 54,643 ha under cultivation and 86,281 MT produced, is consumed in the household as well as for fodder. Oilseeds, with nearly 39,000 MT of output, indicate scope for processing. Pulses, critical for protein nutrition, currently occupy 8843 ha and production is 8,221MT.

Figure 9: Top Non-Perishable Produce producing districts in Arunachal Pradesh.



Analysis by The Infravision Foundation (2025); Source: Data based on Directorate of Economics and Statistics, Department of Agriculture and Farmer Welfare, Ministry of Agriculture and Farmer Welfare

Figure 10 indicates the districts that grow the major non-perishable crops. Major paddy growing districts are Lower Dibang Valley, East Siang and Papum Pare (both eastern and western regions), which have better access to water. Millets and cereals are being grown in districts that are concentrated in the northeastern region of the state. Lower Dibang Valley, Lohit, Namsia, and Chanlang districts have the terrain and climate that is more suitable for hardy grains. Maize cultivation is visible in districts such Lower Dibang Valley, Lohit, Chanlang, East Siang and West Siang. Pulses and oilseeds are notably found in districts of Namsai, Lohit, and Changlang, East Siang and West Siang.

5.2 Item wise description of Non-Perishable Produce: Arunachal Pradesh in comparison with NER

5.2.1 Paddy

Arunachal Pradesh contributes modestly to paddy cultivation in the Northeastern Region (NER), with 1.37 lakh hectares under cultivation and a total production of 2.54 lakh tonnes. Its productivity (1.85 MT/ha) is below the NER average (2.54 MT/ha). While Arunachal outperforms Manipur (1.25 MT/ha) and Mizoram (1.73 MT/ha), it lags significantly behind states like Tripura (3.26 MT/ha), Nagaland (2.63 MT/ha), and Assam (2.62 MT/ha). The productivity gap can be addressed through interventions in irrigation, mechanization, seed quality improvement, and extension services to farmers.

Table 1. Paddy Production and Productivity

States	Area (ha)	Production (MT)	Productivity (MT/ha)
Assam	23,07,992	60,44,899	2.62
Tripura	2,41,798	7,88,822	3.26
Nagaland	2,11,130	5,54,669	2.63
Arunachal Pradesh	1,37,166	2,54,396	1.85
Manipur	1,86,380	2,32,820	1.25
Meghalaya	95,297	2,22,305	2.33
Mizoram	34,071	58,836	1.73
Sikkim	8,190	15,290	1.87
NER total	32,22,024	81,72,037	2.54

Source: Directorate of Economics and Statistics 2022-23, Ministry of Agriculture and Farmer Welfare

5.2.2 Maize

With 54,643 hectares of land under maize cultivation, Arunachal Pradesh ranks second in the region after Assam. However, the productivity in maize remains low (1.58 MT/ha), which is significantly below the NER average (2.63 MT/ha). While Arunachal Pradesh is second to Assam in terms of area under maize cultivation, its productivity is significantly lower compared to Assam (4.06 MT/ha), Tripura (2.34 MT/ha) and Meghalaya (2.27 MT/ha). Efficient agronomic practices and improved crop management can boost maize yield.

Table 2. Maize Production and Productivity

State	Area (ha)	Production (MT)	Productivity (MT/ha)
Assam	1,05,722.00	4,29,003.00	4.06
Nagaland	44,412.00	88,931.00	2.00
Arunachal Pradesh	54,643.00	86,281.00	1.58
Sikkim	37,523.00	67,046.00	1.79

Manipur	25,480.00	56,560.00	2.22
Meghalaya	19,482.00	44,131.00	2.27
Tripura	14,718.00	34,409.00	2.34
Mizoram	11,546.00	19,322.00	1.67
NER Total	3,13,526.00	8,25,683.00	2.63

Source: Directorate of Economics and Statistics 2022-23, Ministry of Agriculture and Farmer Welfare

5.2.3 Pulses¹⁰

Arunachal Pradesh cultivates pulses over 8,843 hectares, yielding 8,221 MT, with a productivity of 0.93 MT/ha. While this is slightly above the NER average of 0.88 MT/ha, the state lags in absolute production due to limited cultivated area. Meghalaya (1.25 MT/ha) and Mizoram (1.22 MT/ha) stand out with the highest productivity in the region, whereas Assam, despite the largest area under pulses (1.24 lakh ha), has a low productivity of just 0.75 MT/ha, pulling down the regional average. While Arunachal has favourable agro-climatic conditions for pulses, its potential remains underutilized. Scaling up the area under cultivation and improving access to quality seeds and extension services can help enhance pulse output, particularly as pulses require relatively low inputs. Since the production is mostly done without using chemical fertilisers and pesticides, the state can claim that pulses produced in the state are organic.

Table 3. Pulses Production and Productivity

States	Area (ha)	Production (MT)	Productivity (MT/ha)
Assam	1,24,867	93,787	0.75
Nagaland	27,085	32,321	1.19
Manipur	32,160	31,680	0.99
Tripura	17,312	13,946	0.81
Arunachal Pradesh	8,843	8,221	0.93
Meghalaya	6,117	7,637	1.25
Mizoram	4,925	6,010	1.22
Sikkim	5,158	5,010	0.97
NER Total	2,26,467	1,98,612	0.88

Source: Directorate of Economics and Statistics 2022-23, Ministry of Agriculture and Farmer Welfare

5.2.4 Oilseeds

Oilseeds produced in the NER include linseed, castor seed, rapeseed, mustard, Niger seed, sunflower, sesamum, groundnut and soyabean. Arunachal Pradesh recorded an area of 36,724 hectares under oilseeds, with a production of 38,919 MT, achieving a productivity of 1.06 MT/ha. This is significantly higher than the NER average of 0.83 MT/ha, indicating efficient land use and relatively favourable growing conditions in the state. Among all states, Meghalaya (1.07 MT/ha) leads marginally in productivity, closely followed by Arunachal, Mizoram, and Nagaland—all above 1.0 MT/ha.

This suggests that Arunachal has an edge in oilseed productivity but needs to expand the cropped area and promote value addition (e.g. through oil extraction units) to enhance income opportunities and reduce the NER's dependence on edible oil imports.

¹⁰ Total pulses include ahar/tur, moong, urad, gram, moth, khesari, cowpea, masoor, peas and beans and other kharif/rabi pulses.

Table 4. Oilseeds Production and Productivity

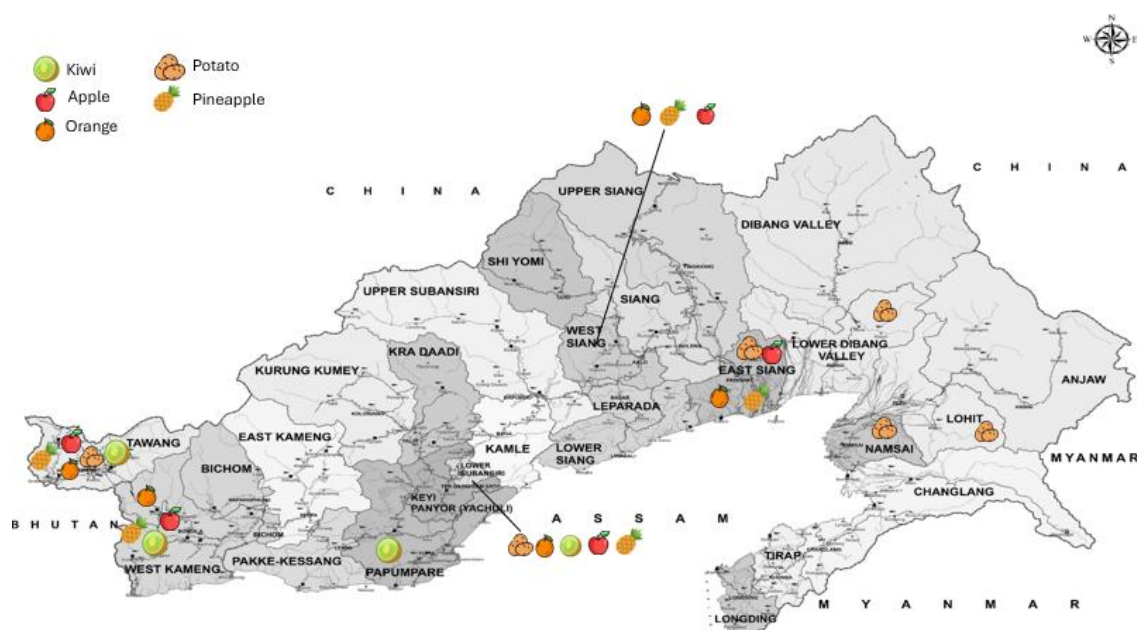
State	Area (ha)	Production (MT)	Productivity (MT/ha)
Assam	3,42,080	2,65,567	0.78
Nagaland	39,500	41,080	1.04
Arunachal Pradesh	36,724	38,919	1.06
Manipur	38,980	29,850	0.77
Meghalaya	14,351	15,339	1.07
Tripura	13,249	11,239	0.85
Sikkim	5,596	5,193	0.93
Mizoram	4,474	4,698	1.05
NER	4,94,954	4,11,885	0.83

Source: Directorate of Economics and Statistics 2022-23, Ministry of Agriculture and Farmer Welfare

5.3 Perishable Produce

Perishable produce refers to food items that spoil, decay, or become unsafe to eat quickly if not stored properly, usually requiring refrigeration or immediate consumption. Examples include fresh fruits, vegetables, meat, poultry, fish, mushrooms, dairy products, etc. These foods have a short shelf life and are sensitive to temperature and handling conditions.

Figure 10: Top Perishable Produce producing districts in Arunachal Pradesh.



Analysis by The Infravision Foundation (2025); Source: Data based on National Horticulture Board, GOI

Figure 11 illustrates the major producing districts of horticultural fruits—kiwi, apple, orange, and pineapple—across Arunachal Pradesh. Kiwi cultivation is largely concentrated in the western and central districts such as Tawang, West Kameng, and Lower Subansiri, benefiting from the temperate climate and hilly terrain suitable for the crop. Other major producing districts for fruits

like apple, pineapple and orange are Tawang, West Kameng, Lower Subansiri, East Siang and West Siang. Orange is the principal crop occupying 64% of the total horticulture area and contributes 70.9% of the total production of horticultural fresh fruits. Pineapple, banana, apple and kiwi occupying 10.5%, 8.2%, 7.07% and 5.1% respectively of the total horticultural area. Strengthening cold chain infrastructure, local processing units, and market access in these key districts could significantly enhance the economic returns from fruit cultivation in the state.

Potato cultivation is scattered across the state, especially in districts like Tawang, Lower Subansiri, and Lower Dibang Valley, Lohit, East Siang and Namsai. Spices like turmeric and large cardamon are grown across all districts. This geographic spread underscores Arunachal Pradesh's strong potential for expanding high-value horticultural production.

High-value crops like potato, ginger, and large cardamom are cultivated in comparatively smaller pockets but offer strong yield performance. For example, potato cultivation spans only 6,502 ha but yields over 48,000 MT. Similarly, ginger, grown over 3,804 ha, generates more than 12,000 MT, demonstrating strong commercial potential. Conversely, large cardamom, though grown over a similar area (6,584 ha), has relatively low output (1,606 MT), indicating the need for targeted intervention in this crop.

The high performance of perishable crops like ginger and potato indicates the potential for promotion of these crops through aggregation of production clusters and investment in post-harvest infrastructure and facilities for cleaning, grading, packaging etc. Cold chain facilities are required at the aggregation clusters so that the produce of farmers and FPOs can be preserved in a scientific manner.

5.4 Item wise description of Perishable Produce: Arunachal Pradesh in comparison with NER

5.4.1 Kiwi

Arunachal Pradesh is the leading kiwi-producing state in India, accounting for nearly 40% of the national production and over 70% of the cultivation area within the NER. While the state devotes the most land area (3,558 Ha) in the NER, its productivity (2 MT/ha) is significantly below both the NER average (3.36 MT/ha) and the national average (3.31 MT/ha). States like Meghalaya (11.22 MT/ha), Manipur (10.58 MT/ha), and Nagaland (5.12 MT/ha) demonstrate much higher productivity levels. Investment in orchard management practices, high-quality planting material, and capacity building for farmers can significantly improve the yield of kiwi fruit. Lower Subansiri, Papum Papre, West Kameng, Tawang are promising districts for kiwi production.

Table 5. Kiwi Production and Productivity

States	Area (ha)	Production (MT)	Productivity (MT/ha)
Arunachal Pradesh	3,558	7,111	2.00
Manipur	286	3,025	10.58
Sikkim	175	1,964	11.22
Meghalaya	300	1,030	3.43
Nagaland	322	1,650	5.12
Mizoram	408	2,173	5.33
NER Total	5,049	16,953	3.36
India Total	5,442	18,014	3.31

5.4.2 Orange

Arunachal Pradesh contributes less than 1% to India's total orange output from the NER. Its productivity is considerably lower than the NER average of 6.49 MT/ha and lags well behind Assam (14.13 MT/ha), Tripura (14.66 MT/ha), and Manipur (9.92 MT/ha). This relatively low performance is likely linked to aging orchards, limited access to quality inputs, and weak post-harvest infrastructure. The common practice of informal exports to Bangladesh where Arunachal's oranges are processed and re-exported further highlights the need for improved local value-addition capabilities, branding, and market linkage systems to capture greater economic returns within the state. Lower Subansiri, West Kameng, Tawang, West Siang, East Siang are promising districts for orange production.

Arunachal Mandarin (Wakro Orange), which has been accorded GI tag status in recognition of its superior quality and vibrant pigmentation, has received inadequate branding and promotion. Targeted and sustained efforts are needed to position the Wakro Orange as a lucrative source of income for the growers.

Table 6. Orange (including the GI tagged Wakro Orange) Production and Productivity

States	Area (ha)	Production (MT)	Productivity (MT/ha)
Assam	14,986	2,11,752	14.13
Arunachal Pradesh	14,495	84,051	5.80
Mizoram	16,667	54,168	3.25
Meghalaya	10,462	50,245	4.80
Manipur	4,838	48,015	9.92
Nagaland	5,707	46,785	8.20
Sikkim	13,267	26,687	2.01
NER Total	80,422	5,21,703	6.49
India TOTAL	4,59,915	61,78,069	13.43

Source: National Horticulture Board 2022-23, GOI

5.4.3 Pineapple

Arunachal Pradesh's contribution to national production of pineapple remains modest at 20,444 MT, which is merely 1.18% of India's total. Productivity in the state is 6.41 MT/ha, significantly lower than the NER average (13.33 MT/ha) and far below the national average (16.92 MT/ha). By comparison, states such as Assam (19.00 MT/ha), Tripura (14.66 MT/ha), and Nagaland (13.58 MT/ha) perform substantially better. By focusing on inputs for high yields and improving post-harvest handling, especially sorting, packaging and cold storage, can contribute to realizing the potential in pineapple cultivation in Arunachal Pradesh. Lower Subansiri, West Kameng, Tawang, West Siang, East Siang are promising districts for pineapple production.

Table 7. Pineapple Production and Productivity

States	Area (ha)	Production (MT)	Productivity (MT/ha)
Assam	18,024	3,42,438	19.00
Tripura	11,698	1,71,488	14.66
Meghalaya	12,606	1,42,813	11.33
Manipur	12,288	1,30,861	10.65
Nagaland	9,037	1,22,748	13.58
Mizoram	5,027	29,023	5.77

Arunachal Pradesh	3,189	20,444	6.41
Sikkim	199	998	5.02
NER Total	72,068	9,60,813	13.33
India Total	1,08,013	18,27,983	16.92

Source: National Horticulture Board 2022-23, GOI

5.4.4 Apple

Apple cultivation in Arunachal Pradesh is relatively limited, with 4,438 hectares yielding 6,825 MT. The productivity of 1.54 MT/ha is below the NER average of 1.84 MT/ha and significantly lower than that of Nagaland (7.35 MT/ha). Compared to the national average of 9.45 MT/ha, Arunachal's poor performance implies significant gaps. Despite having favourable agro-climatic conditions for apple cultivation, the state has not fully leveraged the potential. Deeper analysis of the challenges in apple production will suggest the targeted interventions that are required to boost competitiveness in this high-value horticultural crop. Lower Subansiri, West Kameng, Tawang, West Siang, East Siang are promising districts for apple production.

Table 8. Apple Production and Productivity

States	Area (ha)	Production (MT)	Productivity (MT/ha)
Arunachal Pradesh	4,438	6,825	1.54
Nagaland	241	1,772	7.35
NER Total	4,679	8,597	1.84
India Total	3,04,172	28,75,809	9.45

Source: National Horticulture Board 2022-23, GOI

5.4.5. Potato

Potato productivity in Arunachal Pradesh stands at 7.4 MT/ha, below the NER average of 9.68 MT/ha. While higher than Mizoram (6.34 MT/ha), its productivity trails behind states such as Meghalaya (11.42 MT/ha), Nagaland (11.17 MT/ha), and Assam (9.75 MT/ha). With 6,502 hectares under cultivation and a respectable production volume of 48,146 MT, Arunachal has a potential for commercial potato farming. Investment in cold storage, seed distribution, and better market integration could help unlock greater returns for potato farmers in the state. Lower Subansiri, Tawang, West Siang, East Siang, Lower Dibang Valley, Namsai and Lohit are promising districts for potato production.

Table 9. Potato Production and Productivity

State	Area (ha)	Production (MT)	Productivity (MT/ha)
Assam	99,029	9,65,654	9.75
Meghalaya	13,523	1,54,474	11.42
Manipur	15,000	1,22,810	8.19
Nagaland	5,273	58,912	11.17
Arunachal Pradesh	6,502	48,146	7.40
Mizoram	213	1,350	6.34
NER Total	1,39,540	13,51,346	9.68

Source: Directorate of Economics and Statistics 2022-23, Ministry of Agriculture and Farmer Welfare

5.4.6. Large Cardamom

Arunachal Pradesh is the third-largest producer of large cardamom in India, following Sikkim and Kerala. The state has 6,584 hectares under cultivation, contributing 1,606 MT to national output.

However, its productivity of 0.24 MT/ha is below both the NER average (0.29 MT/ha) and Nagaland's higher yield of 0.56 MT/ha. Arunachal does outperform Manipur (0.02 MT/ha) and is comparable to Sikkim (0.25 MT/ha). Strengthening post-harvest practices such as improved drying, sorting, packaging and branding will be crucial to enhancing both the quality and marketability of large cardamom for domestic and export markets. Large cardamom is cultivated across all districts of Arunachal Pradesh, with Lohit emerging as a particularly promising district in terms of production potential.

Table 10. Large Cardamom Production and Productivity

States	Area (ha)	Production (MT)	Productivity (MT/ha)
Sikkim	22,147	5,626	0.25
Nagaland	4,092	2,274	0.56
Arunachal Pradesh	6,584	1,606	0.24
Manipur	217	5	0.02
NER Total	33,040	9,511	0.29
India Total	86,516	36,212	0.42

Source: National Horticulture Board 2022-23, GOI

5.4.7. Ginger

Arunachal Pradesh achieves a ginger productivity of 3.35 MT/ha, which is lower than the NER average of 7.54 MT/ha and significantly behind Manipur (13.53 MT/ha), Assam (9.49 MT/ha), and Tripura (8.79 MT/ha). There is need for better planting material, disease control, and extension services. Reduction in post-harvest losses and improvements in value-chain logistics especially grading, drying, and milling could help the state derive greater economic value from ginger production.

Table 11. Ginger Production and Productivity

States	Area (ha)	Production (MT)	Productivity (MT/ha)
Arunachal Pradesh	3,804	12,742	3.35
Assam	17,997	1,70,755	9.49
Manipur	4,307	58,283	13.53
Meghalaya	10,764	70,432	6.54
Mizoram	8,553	60,131	7.03
Nagaland	4,962	38,000	7.66
Sikkim	15,277	81,815	5.36
Tripura	2,128	18,704	8.79
NER Total	67,792	5,10,862	7.54
India Total	1,90,959	22,01,187	11.53

Source: National Horticulture Board 2022-23, GOI

Summary

Across key crops in Arunachal Pradesh, infrastructure inefficiencies are a major constraint to improving productivity and value realisation:

1. **Seed and Extension Services (Paddy, Pulses, Maize):** Weak extension systems and poor access to quality seeds hinder efficient cultivation practices and farmer support, particularly for pulses and maize.

2. **Irrigation and Mechanization (Paddy, Maize):** Inadequate irrigation infrastructure and limited mechanization reduce crop yields and productivity, especially for paddy and maize.
3. **Post-Harvest and Cold Storage (Pineapple, Potato, Large Cardamom, Ginger):** Lack of infrastructure for cold storage, drying, grading, sorting, and packaging leads to high post-harvest losses and limits marketability and income generation.
4. **Value Addition and Processing Units (Oilseeds, Ginger, Large Cardamom):** Absence of local processing infrastructure (e.g., oil extraction, milling) restricts income opportunities and forces dependence on imports or external markets.
5. **Orchard and Crop Management (Kiwi, Apple, Orange):** Deficiencies in orchard management infrastructure and lack of investment in planting material and crop-specific support reduce yields and profitability in horticulture.
6. **Market Integration and Branding (Potato, Large Cardamom, Ginger):** Poor market linkages and inadequate branding limit access to higher-value markets, particularly for niche crops like large cardamom and ginger.

Addressing these infrastructure gaps—across irrigation, post-harvest, mechanization, seed systems, and market logistics—is critical to unlocking the agri-economic potential of Arunachal Pradesh.

Apart from the above-mentioned challenges there is need for expansion of the plantation area. A 2017 study by NEDFi – “Study and Action Plan for Promoting Fruits and Vegetable Processing Industries in the state of Arunachal Pradesh” also observed that the current crop production levels in Arunachal Pradesh are insufficient to support large-scale food processing industries focused on any single crop. Among fruits, only pineapple, orange, and apple are produced in quantities that hold commercial promise. Other fruits like kiwi, passion fruit, and walnut show good potential, but their production needs to be scaled up significantly to attract meaningful investments. On the vegetable front, crops such as potato and tomato, along with ginger and large cardamom from the spices category, present opportunity for processing and value addition.

Table 12. The crop-wise potential of area expansion of the focus crops in the state (NEDFi (2017))

Focus Crop	Utilized Area (Ha)	Potential Area (Ha)
Perishable Produce		
Apple	14,070	14,000
Kiwi	3,450	9,500
Orange (Citrus)	39,050	1,09,000
Ginger	7,000	46,000
Pineapple	12,280	27,500
Large Cardamom	12,770	12,500
Potato	4,700	34,500
Non-Perishable Produce		
Walnut	4,810	10,500
Total	98,130	2,63,500

Source: NEDFi Report, 2017

5.5 Work Plan for the identified Agriculture and Horticulture Produce of Arunachal Pradesh

5.5.1 Work Plan for Non-Perishable Crop Clusters in Arunachal Pradesh

Table 13. Work Plan for Non-Perishable Crop Clusters in Arunachal Pradesh

District	Major Crops	Recommended Interventions	Priority Level
Lower Dibang Valley	Paddy, Millets, Cereals, Maize	Cluster-based aggregation centres having facilities for cleaning, grading, packaging, testing and scientific storage. For certain items, multipurpose cold room are required. For seed storage scientific infrastructure is required. For other inputs modern infrastructure is needed at aggregation centres. Primary processing facilities at such centres can provide support to FPOs in creating a modern supply chain which can add value to the produce. <i>Interventions to be determined based on major crops</i>	High (Diversified Hub)
East Siang	Paddy, Maize, Potato, Pulses, and Oilseeds		High
Papum Pare	Paddy, Maize		Medium
Lohit	Millets, Cereals, Maize, Pulses, and Oilseeds		High
Changlang	Millets, Cereals, Maize, Pulses, and Oilseeds		High
Namsai	Millets, Pulses, and Oilseeds		High
West Siang	Maize, Pulses, and Oilseeds		High

Source: Analysis by The Infravision Foundation

Table 13 is suggested Work Plan and would require detailed study and field visits to finalise recommendations.

5.5.2 Work Plan for Perishable Crop Clusters in Arunachal Pradesh

Table 14. Work Plan for Perishable Crop Clusters in Arunachal Pradesh

District	Major Fruits	Recommended Interventions	Priority Level
Tawang	Kiwi, Apple, Orange, Potato	High-quality planting material, orchard management training, Orchard rejuvenation, investment in modern nurseries, cold chain infrastructure, solar-powered cold storage, sorting/grading units, pack houses, fruit processing unit, branding of temperate fruits, market access infrastructure	High (Kiwi & Apple Belt)
West Kameng	Kiwi, Apple, Orange		High (Diversified Horti Hub)
Lower Subansiri	Kiwi, Apple		High
East Siang	Orange, Pineapple, Potato		Medium-High
West Siang	Apple, Orange, Pineapple	Fruit cluster development model, Value addition for citrus, quality input distribution, GI-tag branding (e.g., Wakro Orange)	Medium-High
Tawang	Potato		Medium
Lower Subansiri	Potato		Low-Medium

Source: Analysis by The Infravision Foundation

Table 14 is a suggested Work Plan and would require detailed study and field visits to finalise the Work Plan.

Summary

The work plan for Arunachal Pradesh can prioritize high-diversity crop districts like Lower Dibang Valley, Namsai, Tawang, East and West Kameng, Lower Subansiri and Changlang for cluster-based aggregation centres, integrated clearing, grading, packaging facilities. Targeted crop-specific zones such as Papum Pare (maize) require focused infrastructure like seed storage. Key actions can include enhancing rural logistics, setting up processing units, and developing farmer training and market linkage systems. Some case examples in Arunachal Pradesh on the farming success and challenges are enclosed in **Annexure 2**.

6.1. Organic Agriculture in Arunachal Pradesh: A Strategic Opportunity

6.1.1 Organic Certification and other Government Initiatives

National Programme for Organic Production (NPOP) describes organic farming as *“A method of farming without the use of chemical fertilizers and pesticides with an environmentally and socially responsible approach that works at the grass root level preserving the reproductive and regenerative capacity of the soil, good plant nutrition, and sound soil management, producing nutritious food rich in vitality which has resistance to diseases.”*

India follows two organic certification systems, both aligned with national standards but differing in verification and documentation methods:

- a) The National Programme for Organic Production (NPOP) – for exports; and
- b) The Participatory Guarantee System (PGS-India) – for domestic and local markets.

The NPOP outlines several mandatory criteria that farmers must meet to obtain organic certification¹¹. The key requirements include:

1. **Conversion Period:** Land previously used for conventional farming must transition to organic practices, avoiding synthetic chemicals and GMOs. This transition typically takes 2 to 3 years.
2. **Land Requirements:** The land must be free from prohibited substances and should include buffer zones to avoid contamination from nearby non-organic farms.
3. **Use of Inputs:** Only organic seeds, fertilizers, and pest control substances are allowed. The use of genetically modified organisms (GMOs) or synthetic chemicals disqualifies the certification.
4. **Farm Documentation:** Farmers are required to maintain detailed records of all farm activities, such as seed usage, fertilizer application, and irrigation schedules.
5. **Farm Management Plan:** A comprehensive plan must be prepared to outline all intended organic farming activities, serving as a guideline for implementation.
6. **Product Labelling:** Certified products may bear the Indian Organic logo, provided labelling and packaging comply with NPOP guidelines.
7. **Compliance Post-Certification:** Farmers must follow NPOP standards continuously and undergo regular inspections. Any violation can lead to suspension or cancellation of certification.

¹¹ <https://timesofagriculture.in/how-to-get-organic-certification-in-india/>

The NPOP schema lays out criteria in three broad areas: increasing the area of land and soil that can support organic farming, strictly controlled inputs used for farming the land, and closely monitored management of the farm. Achievement of all three conditions would qualify the product of such farming to be labelled as 'organic'. However, it is not sufficient for a product to be certified as organic. It must also find its way to a market for organic produce. Once the organic product enters the supply chain, it must preserve its organic character and thus requires specialised infrastructure for storage, transportation and delivery to the consumer. In order to fully avail the benefits of the organic certification, Arunachal Pradesh must ensure that the full value chain of organic produce is efficient, affordable and protected from contamination, as well as ensuring that the produce can reach the market and command a competitive price in domestic and overseas markets.

6.1.2 Status of Organic Farming in Arunachal Pradesh

Arunachal Pradesh has emerged as a national frontrunner in organic agriculture, with more than 80% of its cultivated land already under organic practices. The state's commitment to natural farming—relying solely on organic fertilizers, bio-pesticides, and non-GMO seeds—has positioned it as a model for producing sustainable and health-conscious food supplies. According to the USDA Organic Market Report 2021, India's organic food sector is poised for rapid growth, with the NER playing a pivotal role in the transformation.

From 2017 to 2022, Arunachal Pradesh recorded a 20% increase in organically farmed area, driven by proactive government policies and farmers engagement. The Agricultural and Processed Food Products Export Development Authority (APEDA) reports a 50% growth in organic exports from the Northeast between 2018 and 2022 and notes the rising competitiveness of organic produce from the NER in international organic markets.

Specialty products such as dried kiwi, known for their concentrated vitamins, fibre, and minerals, represent the state's potential in high-value organic exports. In addition to kiwi, Arunachal Pradesh boasts of other fruits such as plum, cherries, peach, and walnut as well as mushrooms, spices and aromatic and medicinal plants¹². The state has a rare bounty of geographic conditions, a mix of plains, valleys, and hills that support varied soils and microclimates, enabling the cultivation of a wide range of organic crops while also serving as a natural hub for testing climate-resilient agricultural practices.

A notable indicator of Arunachal's organic commitment is its remarkably low consumption of synthetic inputs. In 2023–24, India's total NPK fertilizer consumption stood at 30.64 million metric tonnes, whereas Arunachal Pradesh consumed 599 metric tonnes – a mere fraction of the national figure – which reinforcing its status as a predominantly organic state. Several favourable factors can be identified with respect to organic farming in Arunachal Pradesh:⁷

1. Rainfed subsistence agriculture
2. Land is virtually chemical-free since crops are grown with low synthetics
3. Integrated farming systems (IFS) are woven into the rural lifestyle
4. The yield gap during the transition from chemical to organic farming is expected to be negligible in Arunachal Pradesh, with 80% of farmland already under organic farming. Niche crops like orange, sticky rice, large cardamom, star anise, kiwi, etc. have high market demand
5. Very high floral and faunal diversity

¹² Avasthe, R., et al. (2021) Prospects and Strategies for Organic Farming in Arunachal Pradesh. Available at <file:///C:/Users/admin/Downloads/ProspectsandStrategiesforOrganicFarminginArunachalPradesh.pdf>

6. Farmers in mid and high mountain areas mostly rely on natural resources available on their own farms.

Arunachal Pradesh government has actively advanced organic farming through key initiatives like the Mission Organic Value Chain Development for the Northeast Region (MOVCD NER), which provides ₹46,500 per hectare in financial support for organic inputs and has resulted in expanding organic cultivation to 8,500 hectares. Over 5,000 farmers have benefited from complementary schemes. Cooperative farming models and community initiatives are reducing input costs and increasing farmer incomes through shared resources and bulk procurement. Skill-building efforts are also strengthening the sector. ICAR-NOFRI, Himalayan University, and Krishi Vigyan Kendras (KVKs) have trained over 4,500 farmers in organic vegetable and spice cultivation, integrated farming, and quality seed production, equipping them with practical, market-relevant skills¹³. With over 80% of its farmland already under organic practices, Arunachal Pradesh is leading India's organic agriculture movement

The surge in organic agriculture in Arunachal Pradesh represents more than a transition in farming, it signals a transformative economic opportunity aligned with global demand for wellness and sustainable products. Targeted and ambitious investment in organic farming in the state can deliver unprecedented economic impact. The state can lead India's organic revolution and emerge as a global exemplar in sustainable, climate-resilient agriculture.

¹³<https://arunachaltimes.in/index.php/2025/02/23/organic-agriculture-in-arunachal-prospects-challenges-and-the-way-forward/>



Photo: Freshly harvested oranges from a farm in Arunachal Pradesh

Part 3. Infrastructure Requirements for Agricultural Produce

7.1 Infrastructure Gap Assessment

Arunachal Pradesh produces a wide range of both perishable and non-perishable agricultural commodities, particularly horticultural crops. These products are typically sold by local farmers to intermediaries or local traders, who then supply them to larger market hubs in Guwahati and in Delhi and other trading centres across India. Many farmers are forced to sell their produce immediately after harvest, often at distressed prices. This limits their access to larger, more remunerative markets and undermines the state's efforts to build sustainable value chains, particularly for export-grade produce.

Although the output of both perishable and non-perishable produce is increasing, the infrastructure gaps reduce the overall productivity and efficiency in the supply chain and limit the extent to which farmers can capture the full value of their crops. Infrastructure gaps are prevalent at different stages of processing and marketing of produce, which highlights the need for an Agri Infrastructure Plan for the state. The key gaps that need to be addressed have been described below.

A major gap for the horticulture sector in Arunachal Pradesh is inefficient evacuation of crops and high wastage due to remote locations of production clusters. Most of the state's 63 rural markets lack essential infrastructure. Furthermore, the state has minimal organized processing capacity, with underdeveloped facilities for fruit pulping, vegetable dehydration, oil extraction, and

warehousing. A detailed breakdown of crop-wise processing potential, district-level infrastructure proposals, and cost estimates is provided in **Annexure 3**.

7.1.1 Critical Infrastructure gaps affecting perishable produce

The absence of farm proximate post-harvest and handling Infrastructure imposes risks on the producers of perishable commodities such as fruits, vegetables, flowers, and spices. They require uninterrupted cold chains — from farm-level storage to transportation and final distribution — to preserve freshness, reduce post-harvest losses and maintain market value. Arunachal Pradesh faces a severe infrastructure deficit in this regard, hindering its ability to effectively commercialize its perishable produce.

Cold Storage

According to the National Horticulture Board, Arunachal Pradesh has only two cold storage facilities with a combined capacity of 6,000 metric tonnes (MT). This is the lowest among Northeastern states and inadequate for boosting production and trade in high-value perishable crops like kiwi, orange, and ginger, etc. In contrast, Assam, the region's most developed state in terms of horticultural logistics, possesses 43 cold storage facilities with a cumulative capacity of 2,06,742 MT. Other states with comparable or smaller agricultural output also have better infrastructure. Tripura operates 14 cold storage facilities with 46,354 MT capacity, while Nagaland and Meghalaya have 8,150 MT and 8,200 MT capacities respectively. Even Mizoram, with just three facilities, exceeds Arunachal's capacity.

Table 15. Cold Storage Infrastructure in Northeast India

S. No.	States/Union Territories	No. of Cold Storage Units	Capacity (MT) (A)
1	Arunachal Pradesh	2	6,000
2	Assam	43	2,06,742
3	Manipur	2	4,500
4	Meghalaya	4	8,200
5	Mizoram	3	4,071
6	Nagaland	5	8150
7	Sikkim	2	2,100
8	Tripura	14	46,354

Source: National Horticulture Board 2009, GOI

Arunachal Pradesh's competitive disadvantage in horticulture production compared to other NER states, despite its significant organic and horticultural potential, can be improved by focusing on cold storage development. Arunachal Pradesh will need to undertake a comprehensive, state-specific assessment of its horticulture infrastructure gaps. As part of this, establishing cold storage facilities in key high-production horticulture zones such as Ziro (Lower Subansiri), Pasighat (East Siang), Aalo (West Siang), Dirang (West Kameng), and Namstering (Tawang) will be critical to reducing post-harvest losses and improving market access.

Transportation

Arunachal Pradesh faces significant transportation shortages in its Agri supply chains. Roadways remain the primary lifeline and poor intra-state connectivity, frequent landslides, and long transit routes via Assam increase both cost and post-harvest losses. Rail and air connectivity are limited, and inland waterways remain underutilized. Sensitive crops such as kiwi and oranges are commonly transported in non-refrigerated vehicles over long distances, leading to quality deterioration and spoilage. The shortage of refrigerated transport remains a critical gap in remote regions like Arunachal Pradesh, where logistical challenges are magnified by difficult terrain and

poor connectivity. Cardamom marketing will also benefit from cold chain infrastructure. A detailed breakdown of existing routes, travel estimates, and infrastructure constraints is provided in **Annexure 4**.

7.1.2 Infrastructure gaps affecting non-perishable produce

Arunachal Pradesh has high rainfall and humidity, therefore produce must be protected from fungal growth, insect infestation, and water ingress. Non-perishable products such as sticky rice (paddy), pulses, grain, spices require specific infrastructure that protects from moisture, pests, and contamination while maintaining product quality over extended storage periods.

The state government has introduced new policies such as the Arunachal Pradesh Horticulture Policy 2025-35 and the New Agriculture Policy 2025-35, emphasizing infrastructure development to enhance productivity. The policies promote increased coverage, climate-resilient and sustainable practices and crop diversification toward high-value varieties¹⁴.

The example of “Construction of FCI Warehouses in Remote Regions of Arunachal Pradesh” is included in **Annexure 2**, exemplifying a case to reduce high transport costs and improve food grain delivery in remote areas. The Food Corporation of India partnered with the Arunachal Pradesh government to construct modern storage godowns in ten logistically challenging districts. This initiative curbed Hill Transport Subsidy expenses but also strengthened the Public Distribution System and enhanced food security across the state.

7.2 Differentiating Infrastructure Needs for Perishable and Non-Perishable Produce

The post-harvest infrastructure required for agricultural produce varies significantly between perishable (e.g., fruits, vegetables, flowers) and non-perishable items (e.g., cereals, pulses, oilseeds) at the crucial stages of harvesting, storage, and processing. As the produce moves from the farm gate to the market, it enters a value chain in which the farmer is usually dependent on third parties for transportation and market linkage. Beyond the stages of production and processing, the infrastructure tends to converge, with a key difference being the atmosphere control required for perishables and the total quality control required for organics.

Table 16. Differentiating Infrastructure Needs for Perishable and Non-Perishable Produce

	Perishable Produce	Non-Perishable Produce
1	Initial Harvesting and Handling Infrastructure	
PURPOSE	Immediate and timely entry of harvested crop into cold chain	Protection from moisture, pests, and contamination
INFRASTRUCTURE REQUIRED	1. Harvest crates and soft-handling tools 2. On-farm pre-cooling units and cold rooms powered by renewable energy wherever possible 3. Packhouses with grading, washing, and sorting facilities for standardization	1. Galvanized iron or aluminium bins (30–50 quintal capacity) for household and FPO-level storage 2. Threshing floors, drying yards, and mechanical dryers for grains and pulses 3. Batch-type grain dryers for individual and community

¹⁴ Arunachal cabinet approves 3 policies to boost agri, horti and tourism sectors: Times of India (5 March 2025) available at <https://timesofindia.indiatimes.com/city/guwahati/arunachal-cabinet-approves-3-policies-to-boost-agri-horti-tourism-sectors/articleshow/118741988.cms>

		use, running on uninterrupted power supply - Mobile seed grading units for remote locations and Fixed community-level seed processing centres with gravity separators and aspirators. - Storage bins and godowns designed for dry, ventilated conditions
CASE EXAMPLE	A fruit cluster for Wakro Orange, with modular packhouses and small sized, mobile cold rooms for field-level handling	Small silos and cleaning, grading and packaging units for rajma and other pulses grown in the Siang Valley
2	Primary Processing and Storage	
PURPOSE	Value-preserving processing and short-term cold storage	Bulk primary processing and long-term storage at ambient temperature
INFRASTRUCTURE REQUIRED	- Pre-cooling facilities at aggregation centres where FPOs can bring their produce - Integrated quick freezing units at locations where perishable items having demand for Individual Quick Freezing (IQF) can be frozen - Fruit pulp processing units, vegetable dehydration units, juice and jam processing plants - At least a few Controlled Atmosphere (CA) cold storage, especially for long-duration storage	- Dry and properly ventilated godowns at block or cluster level with raised floors and rodent-proof design. - Sorting, grading and packaging units for non-perishable items such as pulses, maize and rice - Modern warehousing with fumigation systems and other regulatory features, such as Warehouse Receipt Systems (WRS) linked to credit for FPOs and SHGs.
CASE EXAMPLE	Establishment of a fruit processing unit for kiwi in Ziro Valley, combined with a CA storage unit	A crop-based processing unit for oilseeds in Lohit district – including oil expelling and seed packaging units
3	Transport and Market Linkage Infrastructure	
PURPOSE	Linking farmers to markets through integrated supply chains that prevent wastage in all commodities and prevent spoilage and preserve freshness in perishables	
COMMON INFRASTRUCTURE REQUIRED for perishable and non-perishable	- Aggregation hubs and regulated warehouses where FPOs can bring their produce - All-weather rural roads and last-mile connectivity to aggregation centres - Storage facilities at aggregation centres - Transportation terminals and transit facilities with loading and unloading facilities - Digital platforms to access and manage logistics infrastructure	

	6. Integration of FPOs to achieve sustainable economies of scale
ADDITIONAL INFRASTRUCTURE FOR PERISHABLES	7. Reefer trucks and cold chain logistics integrated with packhouses and markets 8. Cold storage points at railheads, ports and air cargo terminals
CASE EXAMPLE	A cold chain corridor serviced by reefer trucks, connecting vegetable growers in Dirang to markets in Guwahati, Kolkata and Delhi

To unlock the full potential of both perishable and non-perishable value chains, stage-specific infrastructure planning is essential. While initial stages demand differentiated interventions tailored to perishability, the mid-to-end supply chain infrastructure can be developed in an integrated manner with necessary provision for atmosphere control for perishable produce. The state can adopt cluster-based infrastructure upgradation to address both types of produce and to leverage economies of scale. A multimodal logistics strategy is urgently needed for investment in all-weather rural roads, reefer transport, cold-chain facilities, and packhouses at nodal points. A multimodal logistics strategy is urgently needed to enable seamless market access and reduce spoilage—including investment in all-weather rural roads, reefer transport, cold-chain facilities, and packhouses at nodal points.



Photo: Mithun (*Bos frontalis*) livestock grazing in a valley of Arunachal Pradesh

PART 4 Recommendations

8.1 Recommendations for Agricultural Infrastructure Development in Arunachal Pradesh

Lack of infrastructure and related services for storage, processing and transportation of perishable and non-perishable commodities is a binding constraint on the growth of agricultural production in Arunachal Pradesh. Based on a comprehensive assessment of the state's infrastructure needs and gaps, there are clearly identified areas for policy and investment. The targeted approach is necessary in a cash-strapped and resource-scarce economy.

It is anticipated that the state government will receive support from organisations such as APEDA and will aim to take advantage of the opportunities presented by the entire range of schemes and missions of the Government of India to achieve its goals.

8.1.1 Articulate clear goals and objectives

The goal of planning, investment and management for the agricultural sector in Arunachal Pradesh is to enhance the farmer's income and contribute to prosperity in a state that has great cultural and strategic significance for India. Improving productivity and efficiency in the state's

agricultural value chain and reducing losses and wastage are necessary conditions for achieving the goal.

8.1.1.1 Increasing production and reducing post-harvest losses in perishable commodities

Comprehensive cold chain networks are needed to serve the key horticultural clusters like Lower Dibang Valley, East Siang, and Namsai. Infrastructure networks that combine pre-cooling units at farm gates, distributed cold rooms for farm scale or cluster scale storage, and refrigerated transport systems are critical for high-value crops like kiwi, oranges, and other perishables that currently suffer significant post-harvest losses. The significance of networks is experienced during localised stress caused by weather conditions or infrastructure damage in one district, allowing for alternative solutions, thereby reducing the risk of total crop wastage or spoilage. The aspects requiring priority attention are given below:

1. Expand cold storage capacity to fill the existing 20% gap by investing in decentralized cold storage units at the block or cluster level.
2. Ensure availability of different sizes of reefer vehicles and make them accessible through digital platforms that can use advanced intelligence and data to prioritise availability during the peak harvest seasons and deploy the vehicles in different locations as per harvesting patterns.
3. Connect the FPOs and producer cooperatives to integrated cold-chain infrastructure for greater financial sustainability.
4. Promote public-private partnerships to improve capacity and improve efficiency.

8.1.1.2 Improving processing, storage and transportation for Non-Perishable commodities

Develop a network of community-level threshing floors, drying yards, and mechanical dryers, particularly for important crops like sticky rice, turmeric, and large cardamom. Implement climate-appropriate storage solutions including galvanized iron bins, ventilated godowns with raised floors, and moisture-resistant storage facilities tailored to the high-humidity environment of the state.

8.1.2 Undertake strategic planning of investments and institutional support

The state must take stock of the status of agricultural production with a focus on horticulture produce and a targeting of high-value crops such as kiwi fruit. Data availability is a constraint to planning, and this must be addressed through a frequent and sustained monitoring of production cycles, trends and needs in all districts.

Strategic investments need to be planned at a statewide level.

To illustrate the nature of strategic planning required for all commodities, a detailed and holistic assessment of the kiwi value chain has been included in **Annexure 1**.

8.1.2.1 *Identification of specialized zones and clusters*

1. **Cluster-Based Development:** Promote Crop-Specific Clusters and implement crop-based cluster models that integrate the entire value chain—from production to processing and marketing. For example, a kiwi cluster in Ziro Valley, a ginger cluster in East Siang, or a pineapple cluster in Lohit can help drive economies of scale, improve access to shared infrastructure, and attract investment in storage, processing, and branding.
2. **Farm-Gate Collection and Aggregation Centres:** Set up strategically located farm produce aggregation centres equipped with facilities for sorting, grading, and packing, preferably with integrated pack-house infrastructure. Locating these centres close to production areas will reduce farmers' transport burdens, lower post-harvest losses, and minimize logistics costs. The use of reusable open crates for transporting produce from farms to the aggregation centres will further reduce handling damage. Proper post-harvest handling and packing will enhance the overall quality and market value of the produce.
3. **District-Specific Processing Units:** Establish processing infrastructure aligned with regional production strengths such as the Naara Aaba kiwi wine processing unit in Ziro valley and oilseed processing cluster in Lohit district for sesame and mustard. Processing facilities should incorporate quality testing, packaging, and branding capabilities to enhance market value.
4. **Multi-Commodity Aggregation Centres:** Implement the proposed network of aggregation centres identified by NEDFi, focusing on locations like Namstering (Tawang), Dirang (West Kameng), Pasighat (East Siang), Roing (Lower Dibang Valley), and Ziro (Lower Subansiri). Equip these centres with crop-specific handling facilities including sorting, grading, pre-cooling capabilities, and specialized processing equipment like walnut dryers and cardamom dryers.
5. **Agri-Logistics Parks and Food Processing Zones:** Develop integrated agri-logistics parks at strategic locations that combine storage, processing, packaging, and transport services under one roof. These parks should serve as nodes connecting farm production with market access and include modern warehousing with fumigation systems.

8.1.2.2 *Planning the basic infrastructure to connect zones and clusters*

Power supply networks

Reliable power supply is critical, particularly for cold chains and processing units. Decentralized renewable energy solutions must be prioritized to deal with the state's remote location and expanse of off-grid farms. Reliable and uninterrupted power supply – whether sourced from grid or off-grid energy sources -- are essential for cold storage facilities, processing units, and mechanical dryers, particularly in remote areas.

Irrigation systems

Improved irrigation systems are essential to reduce dependence on erratic rainfall and to improve crop productivity across seasons, especially in hilly terrains where traditional irrigation methods are insufficient. However, these systems need to be planned at different scales, from state to district to clusters and individual farms. The needs assessments must be aggregated from farm

level to larger scales, while the investments in infrastructure and operations need to be planned from larger to smaller scales.

Transportation and logistics networks

1. **Develop all-weather rural roads:** Connect production clusters to aggregation centres and improve existing highways like NH-229 and NH-52 that serve critical agricultural regions. Create strategically located aggregation hubs with loading/unloading facilities along major transport corridors to facilitate smoother connections to markets in Assam and beyond.
2. **Improve Last-Mile Connectivity:** Upgrade and maintain last-mile roads connecting farms to local mandis, aggregation centres, and processing hubs. Many farmlands in remote areas lack reliable access to all-weather roads, which hampers timely and cost-effective transport of produce.
3. **Develop Air Cargo Infrastructure:** Establish cargo handling facilities at Donyi Polo Airport and integrate air freight stations in key hubs to facilitate rapid movement of high-value, perishable products to distant domestic and export markets.
4. **Accelerate Rail Link Projects and Improve Intermodal Transport:** Fast-track the construction of proposed rail lines in the region and enhance coordination between road and rail logistics. This will support bulk movement of Agri-produce, reduce transport costs, and improve market access.

8.1.3 Sustain institutional support to farmers

1. **Ensure that training and skill-building support is freely available:** Create dedicated infrastructure for farmer training on post-harvest management, processing techniques, and market linkages. Support and strengthen existing FPOs and cooperatives with infrastructure that enables collective marketing and shared resource utilization.
1. **Establish soil testing laboratories:** Expand the network of accessible and affordable soil testing facilities across the region to promote scientific nutrient management. This will help improve productivity and address issues of soil health degradation due to unbalanced input use.
2. **Establish certified nurseries:** Support the establishment and upgrading of certified nurseries to ensure the availability of quality planting material. Encourage public-private partnerships and capacity-building programs for nursery operators to maintain varietal purity and plant health standards.
3. **Establish a network of model orchards:** Develop a network of model orchards within key production clusters to demonstrate best practices in horticultural production. These orchards should serve as knowledge hubs, integrating efforts of progressive farmers, scientists, and technical experts. The model orchards will adopt advanced orchard management techniques, including the following:
 - a. Integrated Pest Management (IPM) and Integrated Nutrient Management (INM)
 - b. Adoption of drip irrigation and organic farming methods
 - c. Scientific training and pruning, thinning practices
 - d. Use of growth regulators to enhance fruit size and uniformity

- e. Ongoing capacity-building and technical support

8.1.4 Improve market access and facilitation

1. **Establish food testing and certification facilities:** Set up FSSAI/NABL-accredited food testing laboratories and district-level certification facilities across Arunachal Pradesh to ensure compliance with FSSAI, AGMARK, and organic standards—enabling quality assurance, value addition, and greater access to premium domestic and export markets.
2. **Ensure traceability of produce:** Traceability systems that use barcode or QR-codes to tag the source of the produce and track its movement are needed to ensure end-to-end transparency across the supply chain. Improving buyer confidence and facilitating product recall are essential for gaining access to high-end export markets.
3. **Invest in branding and certification:** Support branding initiatives for NER's unique Agri Horti products, including Geographical Indication (GI) tagging, quality certification, and promotional campaigns. A strong regional brand identity will help build market recognition, increase consumer trust, and unlock premium pricing in both domestic and international markets. A case example of Innovation in branding and marketing in Mumbai is enclosed at **Annexure 3**.

Summary

Arunachal Pradesh faces major infrastructure gaps in storage, processing, and transportation, limiting agricultural export, especially for perishable goods. The state must focus on expanding cold chains, setting up farm-gate aggregation centers, and promoting cluster-based development tailored to high-value crops. Strategic investments in logistics, power, irrigation, and processing infrastructure—combined with improved data collection—are crucial for linking production zones with markets. Institutional support through farmer training, model orchards, certified nurseries, and testing facilities will help improve productivity and product quality. Finally, strengthening market access through food certification, traceability systems, and branding will help farmers reach premium markets and increase incomes.



Photo: Orange orchards in Arunachal Pradesh

Annexure 1

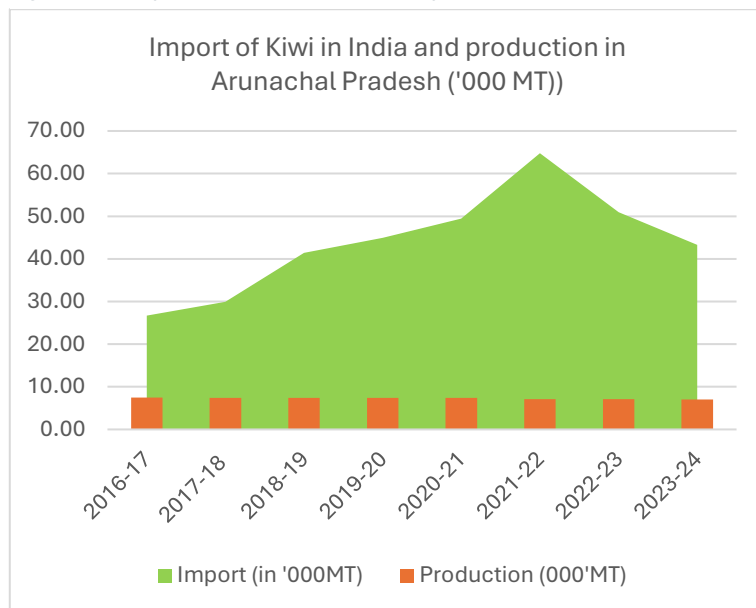
Kiwi: Supply chain and Logistics Infrastructure

Kiwi is an emerging high-value horticultural crop in India, with increasing consumer demand and limited domestic production. Arunachal Pradesh, by virtue of its temperate climate and organic cultivation practices, has emerged as the leading kiwi-growing state, yet remains far from realizing its full commercial potential. This section maps the state's kiwi supply chain and identifies infrastructure gaps holding back sectoral growth.

Kiwi import and production trends:

The data in **Figure 12** shows a widening gap between domestic kiwi production and import demand in India. Over the eight-year period from 2016–17 to 2023–24, India's kiwi imports nearly doubled, rising from 26.69 thousand metric tonnes (TMT) to 43.27 TMT, peaking at 64.78 TMT in 2021–22. This steady growth in imports signals a sharp increase in consumer demand, possibly driven by rising health consciousness and changing dietary patterns.

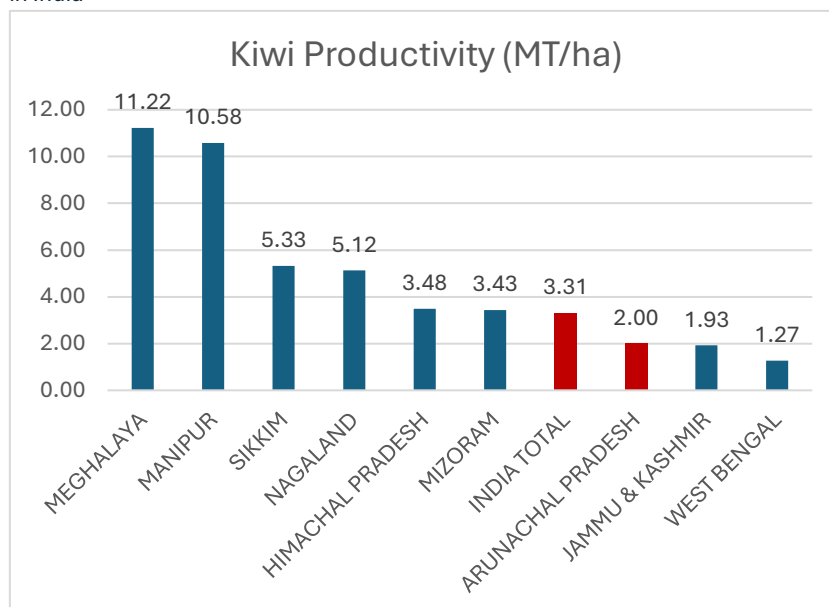
Figure 11. Import of Kiwi in India and production trends in Arunachal



Source: National Horticulture Board 2022-23, GOI

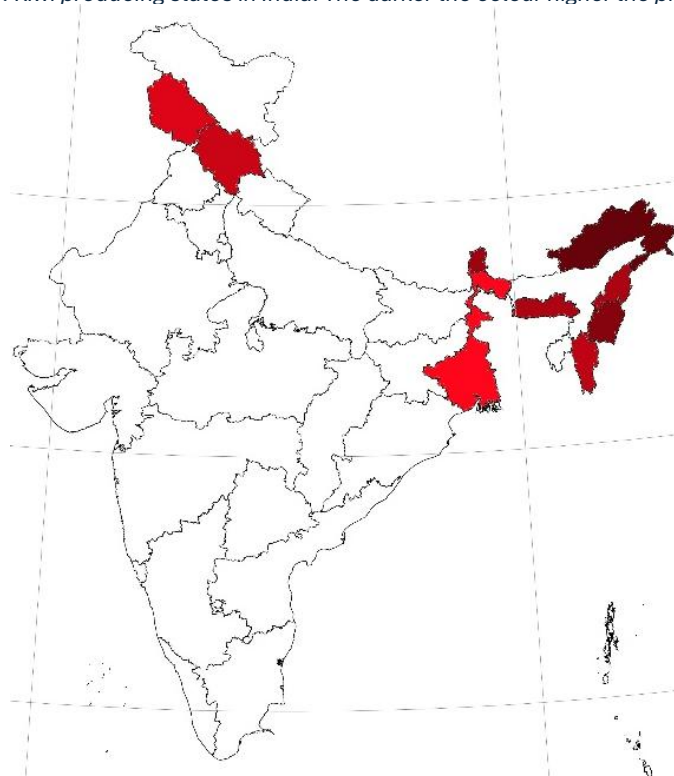
In contrast, domestic kiwi production in Arunachal Pradesh (which contributes ~40% to national production) has remained stagnant, ranging narrowly from 7.45 TMT in 2016–17 to 7.05 TMT in 2023–24, indicating no significant expansion in cultivated area or yield improvement. This mismatch between import dependency and domestic stagnation points to an urgent need for investment in kiwi value chains within India.

Figure 12. Kiwi productivity comparison among kiwi producing states in India



Source: National Horticulture Board 2022-23, GOI

Figure 13. Kiwi producing states in India. The darker the colour higher the production



Analysis by The Infravision Foundation (2025); Source: Data based on National Horticulture Board, GOI

Kiwi cultivation in India is confined to a handful of states (**Figures 13 and 14**) that possess the necessary temperate climate and hilly terrain required for the fruit's growth. The data above highlights production performance across these limited regions, revealing significant inter-state variations in yield and output. Arunachal Pradesh emerges as the largest contributor in terms of area under kiwi cultivation, with 3,558 hectares dedicated to the crop accounting for nearly 65% of India's total kiwi-growing land. However, despite its scale, the state's productivity is notably low at just 2.00 MT/ha, falling significantly below the national average of 3.31 MT/ha. This suggests that the region's full potential is yet to be realized, and targeted improvements in orchard management, post-harvest handling, and agronomic practices are essential to enhance efficiency.

In contrast, states like Meghalaya and Manipur demonstrate exceptional productivity at 11.22 MT/ha and 10.58 MT/ha, respectively, more than three times the national average. Sikkim and Nagaland also show promising results, with productivity figures of 5.33 MT/ha and 5.12 MT/ha respectively, while Mizoram and Himachal Pradesh trail slightly behind with yields around 3.4–3.5 MT/ha. At the lower end of the spectrum are Jammu & Kashmir (1.93 MT/ha) and West Bengal (1.27 MT/ha), which, having smaller cultivation areas, demonstrate limited output.

Overall, this data reinforces the fact that kiwi is a niche crop in India, grown only in select states due to ecological constraints. While Arunachal Pradesh leads in terms of area and absolute production, productivity remains a challenge. There is substantial opportunity to improve yield levels through targeted interventions—especially since the global demand for kiwi continues to rise, and India remains heavily reliant on imports to meet domestic needs.

Supply chain of Kiwi in Arunachal Pradesh:

Arunachal Pradesh, particularly the Lower Subansiri district, has emerged as a significant producer of kiwi fruit in India. The district functions as the major production cluster, growing popular varieties such as Hayward, Monty, Allison, and Bruno. (Hayward dominates market demand due to its larger size and consistent shape) The kiwi harvesting season begins in October and peaks between November and mid-December. However, despite the region's favourable agro-climatic conditions, several infrastructure gaps persist across the supply chain, which limit both value realization for farmers and the long-term competitiveness of kiwi farming in the region.

A report by CSS Institute of Agricultural Marketing mapped out the kiwi supply chain in Arunachal Pradesh. The illustrated that kiwi supply chain begins at the farm level, where farmers handle the production, harvesting, cleaning, grading, and packing of the fruit. From there, the produce is sold to local and distant traders, who are responsible for transporting the fresh kiwis to markets. A portion of the harvest is used for processing. Most of the kiwi reaches consumers through retailers, who sell it in fresh form. Post-harvest losses are primarily attributed to the use of rudimentary packaging methods, such as simple pouches or cartons, which offer limited protection during handling and transportation.

1. Planting Material

One of the most critical gaps lies in the supply of planting material. The three government nurseries in Lower Subansiri are insufficient to meet regional demand. As a result, many farmers rely on informal and often unverified sources for planting material. This leads to a lack of varietal uniformity and an overplanting of male kiwi vines, which do not bear fruit. Orchards end up with mixed varieties, complicating grading, packing, and marketing for high-end buyers who demand uniform produce.

The absence of structured soil and nutrient management further exacerbates productivity issues. Most farmers rely on organic farming and avoid synthetic fertilizers and pesticides. Farmyard manure (FYM) is either bought from the local market or prepared by farmers using cow dung, poultry, and piggery waste. However, the quantity and quality of manure applied vary widely among farmers. Consequently, yields remain low, and fruit sizes are often suboptimal. An estimated 70% of the harvest falls into lower grades (B, C, or D), leading to lower price realization for producers.

2. Garding and Sorting and Processing Units

Post-harvest infrastructure remains severely underdeveloped. Sorting and grading are carried out manually at the farm level based largely on fruit size and taste. Although the fruits are categorized into grades A (>70 g), B (50–70 g), and C (<50 g), most grading is done visually and without proper standardization. Packaging is basic and often substandard, with fruits packed into second-hand cardboard cartons or low-quality plastic crates. This increases the risk of bruising and spoilage during handling and transportation.

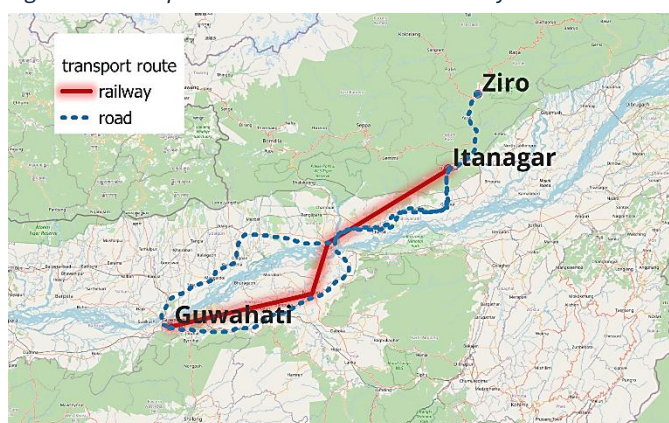
Mechanization and value addition infrastructure are absent in the region. The region lacks cold rooms, integrated grading lines, and mechanized packing facilities. All processes remain labour-intensive, with little access to technical training or standardization. As a result, traders receiving produce at distant markets often need to re-sort and re-grade the fruits, incurring additional costs that are ultimately deducted from the farmer's income.

The **lack of aggregation centres and cold chain facilities** is particularly problematic. There is lack cold storage available at the farm level and lack of provision for pre-cooling or refrigerated transportation. Without these, the kiwi supply chain in Arunachal Pradesh is highly vulnerable to post-harvest losses, especially given the long distances and rough terrain involved in transport.

3. Transport

Transport from the Ziro valley to major market hubs like Guwahati involves long-distance haulage over poor road networks. Kiwis are typically transported in non-refrigerated mini-vans or small trucks, which further contributes to spoilage and quality deterioration during transit.

Figure 14. Transport route for kiwi from Ziro valley to Guwahati



This map (**Figure 15**) shows the transport route of kiwi from Ziro, the main production cluster in Arunachal Pradesh, to Guwahati via Itanagar. Kiwis are first transported by road from Ziro to Itanagar, and then either continue by road or rail to Guwahati or Lakhimpur in Assam, a major trade hub. It can take 12-16 hours (500km) to transport from Ziro valley to Guwahati.

Transport costs are also a significant burden on producers. The cost per truck is approximately Rs. 1,000 to the local Ziro market, Rs. 6,000 to Itanagar, and Rs. 20,000 to Guwahati. These high costs, combined with quality degradation during transit, severely impact the profit margins for kiwi growers.

4. Marketing Channels

The marketing and distribution system for kiwi in Arunachal Pradesh remains largely unstructured. Bulk sale of kiwi does not take place through regulated platforms such as APMC mandis. Instead, the bulk of the produce bypasses formal marketing systems and is channelled through informal routes.

Two predominant supply chain models are observed in the kiwi value chain:

1. Farmer → Commission Agent → Wholesaler → Retailer

This is the most common route, where the farmer sells the produce to a local commission agent. The agent, in turn, connects with wholesale buyers in distant cities, who distribute the fruit through regular retail channels.

2. Farmer → Commission Agent → Wine Processing Unit → Wholesaler → Retailer

In this model, a portion of the kiwi harvest is directed toward wine production units.

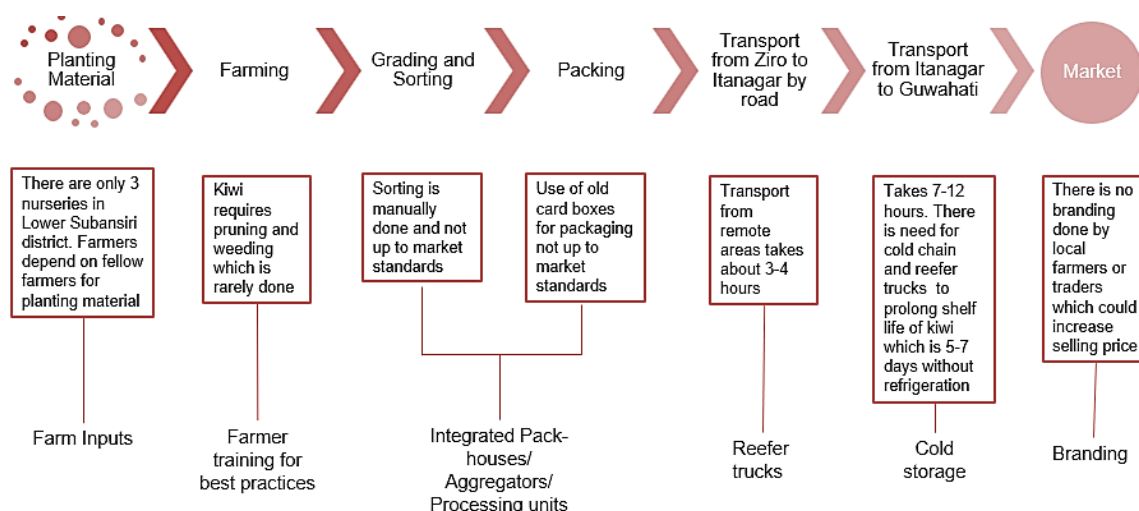
Most of the produce is procured directly at the farm-gate by middlemen or commission agents, limiting the bargaining power of farmers. Only a small group of farmers transport their produce

directly to wholesale markets in cities such as Itanagar, Guwahati, Delhi, and Kolkata, where they negotiate with wholesalers or pre-harvest contractors. However, such direct-to-market linkages remain rare due to logistical constraints, high transport costs, and lack of storage facilities.

An organic wine processing facility, M/s Lambu Subu Food & Beverages, located in Ziro, produces the renowned kiwi wine brand “NAARA-AABA. Approximately, 15% of the produce goes through this channel.

This fragmented and intermediary-heavy marketing structure restricts price realization for farmers and hinders the development of a robust kiwi value chain. Establishing organized aggregation centres, farmer-producer organizations (FPOs), and market linkages can help streamline the marketing process, reduce wastage, and enhance farmer incomes.

Figure 15. Map of Infrastructure gaps in kiwi supply chain



Identified Infrastructure Gaps

Based on the supply chain mapping, the following critical infrastructure gaps have been identified:

1. **Nursery and Manure:** The limited availability of grafted, high-quality planting material and lack of varietal uniformity hinder productivity and market consistency.
2. **Absence of Pack Houses and Aggregation Centres:** The state lacks modern pack houses and grading facilities, leading to inconsistent sorting, poor packaging, and significant post-harvest losses.
3. **Cold Chain Gaps:** There is an absence of on-farm cold rooms, reefer trucks, and ripening chambers, resulting in rapid spoilage of kiwi during storage and transit.
4. **Poor Transport Connectivity:** Weak road infrastructure and long, costly routes without atmosphere controlled reduce produce quality and market competitiveness.
5. **Branding Deficit:** Arunachal’s kiwi lacks geographic branding or collective marketing, (organic) limiting its recognition and premium pricing in competitive markets.

In summary, although Arunachal Pradesh has high potential as a kiwi-producing state due to its favourable climate and organic practices, its fragmented and underdeveloped infrastructure across the supply chain presents a major barrier to market expansion and farmer income growth. Investments in planting material development, post-harvest management, cold chain logistics, and transport connectivity are essential to unlocking the full potential of the kiwi sector in the state.

Annexure 2

Case examples of successes and challenges in Yak churpi, kiwi wine, organic and mushroom farming in Arunachal Pradesh

Case 1: Yak Cheese and the Brokpa Pastoralists of Arunachal Pradesh

In the high-altitude districts of West Kameng and Tawang in Arunachal Pradesh, yak cheese, locally known as *chhurpi*, is both a traditional food and a growing economic opportunity for the Brokpa community, a sub-tribe of the Monpa people. The Brokpas are pastoralists, migrating seasonally with their yak herds to alpine pastures as high as 14,500 feet. This way of life, while ecologically sustainable, is economically precarious, especially as climate change and modern aspirations draw younger generations away from herding.

Yak milk is central to the Brokpa diet and culture. It is processed into a range of products, including *chhurpi* (soft and hard cheese), *churkam* (a type of dried cheese), ghee, curd, and butter. *Chhurpi*, especially the hard, dried variety called *churtang*, is prized for its long shelf life and is a staple during the long migratory periods when fresh food is scarce. Traditionally, these products were also used as barter and as payment for grazing rights in community or monastery-owned pastures.

The market for yak cheese and ghee has grown in recent years, particularly in the towns and tourist markets of West Kameng and Tawang. *Chhurpi* sells for ₹600–₹650 per kilogram and ghee retails at ₹600 per kilogram. However, the supply remains limited due to the relatively low milk yield of these yaks—about 0.98–1.04 kg per day over a 180-day lactation period. Despite this, yak milk is nutritionally superior to cow's milk, being richer in protein, fat, and essential micronutrients.

Traditional cheese-making methods are labour-intensive and rely on indigenous knowledge. The process involves fermenting skimmed yak milk with curd inoculum, churning out butter, and drying the remaining soft cheese into hard *chhurpi*, often stored in yak skin bags. While these traditional methods ensure food security and cultural continuity, they pose challenges for hygiene and scalability, limiting wider market acceptance.

Recognizing the potential of yak milk products, the ICAR-National Research Centre on Yak (NRC-Yak) has introduced interventions such as training Brokpa herders in hygienic processing, quality control, and packaging. The establishment of a yak milk parlour at Nyukmadung and the recent Geographical Indication (GI) tag for Arunachal Pradesh yak *chhurpi* have further boosted the product's profile. NRC-Yak has also developed a semi-sedentary model of yak rearing, complete feed blocks, and silage-making techniques to ensure year-round milk production and reduce the need for arduous seasonal migrations.

Despite these advances, challenges remain. The younger Brokpa generation is increasingly pursuing formal education and alternative livelihoods, leading to a decline in traditional yak herding. The economic viability of yak pastoralism now hinges on value addition, improved infrastructure, and the development of niche markets for high-value yak milk products. There is significant untapped potential in the growing demand for high-value, high-nutrient cheese in both domestic and international markets, which presents a strategic opportunity for Arunachal Pradesh to position itself as a unique supplier of artisanal yak cheese.

In summary, yak cheese production in Arunachal Pradesh is a testament to the resilience and adaptability of the Brokpa pastoralists. With institutional support, market innovation, and the preservation of traditional knowledge, yak cheese is not only sustaining a unique way of life but

also offering a model for the economic revitalization of pastoral communities in the Eastern Himalayas.

Source: <https://scroll.in/article/1063083/in-arunachal-pradesh-growing-market-for-yak-milk-products-helps-sustain-pastoral-community>

Case 2: Organic Kiwi wine from Ziro Valley

Naara-Aaba, located in Hong Village of Ziro Valley, Arunachal Pradesh, is India's first organic kiwi wine producer and a pioneering example of value addition in the region's horticulture sector. Founded in 2017 by agricultural engineer-turned-entrepreneur Tage Rita, the winery was established in response to the recurring problem of kiwi fruit wastage in Ziro, where surplus kiwis often went unsold or were fed to animals due to lack of market access and storage infrastructure.

The winery sources organic kiwis from local farmers and processes them using advanced winemaking techniques and imported machinery to ensure quality and retain the fruit's nutritional value. The production process involves several steps: weighing, cleaning, crushing, fermentation, racking, blending, fining, filtration, cold stabilization, bottling, capping, labeling, and storage.

The enterprise has had a notable impact on the local economy by providing a reliable market and better prices for kiwi farmers, generating employment and inspiring rural entrepreneurship. Rita's leadership and the success of Naara-Aaba have encouraged more locals to consider commercial fruit cultivation and value addition, while also positioning Ziro Valley as a potential wine tourism destination.

Despite its achievements, Naara-Aaba faces significant challenges. Distribution is limited by Arunachal Pradesh's remote location, poor road infrastructure, and market competition from national brands. The lack of cold storage and packhouses increases post-harvest losses, and there is limited awareness of kiwi wine among Indian consumers.

Naara-Aaba's story demonstrates the potential of innovative, organic value addition in the Northeast, combining local resources, women's entrepreneurship, and community development. Its journey from surplus fruit to a nationally recognized wine brand is now cited as a case study by global universities and has brought national attention to Arunachal Pradesh's horticultural production.

Case 3: Organic farming in Upper Siang, Arunachal Pradesh

The organic farming revolution in Upper Siang district is a remarkable story of how a small group of women from Yingkiong circle, inspired by their courage and spirit to learn about organic farming, have helped to create more awareness on organic farming and the ill effects of inorganic farming.

Today, with their teamwork and awareness programmes on organic farming, more people have joined hands, making it the dawn of a revolution in organic farming in Upper Siang.

Totok Libang, a state gold medallist for best performance in agriculture and allied sector, is an inspiration to many. He follows the integrated farming system, which includes paddy-cum-fish culture and vegetable cultivation, and he produces his own vermicompost, as well. He uses solar power to dry the vegetables and fruits, in order to preserve them for a longer time. He also grows organic green tea and wants to acquire organic certification.

Opet Miyu has been growing kiwi for the last eight years. However, selling all her products locally does not fetch her as much income as the kiwis from Ziro Valley in Lower Subansiri district, as

she does not have organic certification. According to her, there is enough space to grow and increase the cultivation “but the farmers are mostly unaware.”

Ampi Liper and Gyan Boli grow orange, pineapple, pearl onion and other horticulture crops whose cultivation is highly productive. Pearl onion, locally known as ‘dilap’, is one of the vegetables commonly grown in Upper Siang. It is very easy to cultivate, and it has certain properties which make it immune to pest attacks. The pearl onion found here is much bigger and better tasting than those found in other parts of Arunachal Pradesh.

The organic farming revolution in Upper Siang district is a shining example of how sustainable agriculture can be a win-win for everyone involved. Farmers benefit from increased income and better health, consumers benefit from access to safe and nutritious food, and the environment is protected from the harmful effects of synthetic pesticides and fertilisers.

Source: <https://arunachaltimes.in/index.php/2023/09/26/organic-farming-revolution-in-upper-siang/>

Case 4: Mushroom Farmer from Namsai District

Lachit Thamoung, a progressive mushroom farmer from Juna IV village in Namsai district, Arunachal Pradesh, was honoured with the ‘Millionaire Farmer of India District Award’. The award ceremony, organized by Krishi Jagran in collaboration with ICAR, was organized from 1st-3rd December 2024, at the ICAR Pusa Mela Ground. The event recognized million-dollar farmers worldwide, with nominations invited from all Indian districts through ICAR institutes. Thamoung was nominated by ICAR-KVK, Namsai.

Starting with a small mushroom unit producing just 10 kg of spawn, Lachit now operates a unit that produces 300 kg of mushrooms annually, including Milky and Oyster varieties. His processing unit is certified by the Food Safety and Standards Authority of India for producing value-added mushroom and meat products. Lachit is also committed to empowering local youth and women, offering training and resources to help them set up mushroom farms.

This recognition highlights Lachit’s dedication to scientific farming methods, agricultural innovations, and his efforts to inspire and mentor others in the farming community. His award serves as motivation for him and others to embrace entrepreneurial ventures in agriculture.

Source: <https://agrospectrumindia.com/2024/12/11/mushroom-farmer-from-arunachal-pradesh-receives-millionaire-farmer-of-india-district-award.html>

Annexure 3

Primary Processing and Storage Facilities

Arunachal Pradesh has six wholesale and 63 Rural Primary Markets, but most lack essential infrastructure. These markets only have basic sheds and minimal drainage, with no facilities for storage, sorting, grading, or packaging. Traditional weighing methods and the absence of price discovery systems lead to post-harvest losses and reduced value for farmers, especially in horticulture.

To improve this, the state must modernize its wholesale markets with cold storage, washing, grading, and packaging units, electronic weighing scales, proper drainage, auction platforms, and kiosks for price information. Facilities for agri-input sales and extension services should also be included. While Rs. 208 lakhs have been sanctioned under Technology Mini Mission–III, an additional Rs. 10 crore is proposed. The state can leverage schemes like MIDH, RKVY, AMI (NABARD), and DoNER to fund this upgrade and build a stronger agricultural marketing system.

Arunachal Pradesh lacks even basic primary processing infrastructure for cleaning, drying, and minimal packaging of agricultural produce¹⁵. This results in a loss of value and limits the quality competitiveness of the products in external markets. The government is investing in infrastructure improvements, such as the establishment of a CSIR-CMERI centre in Naharlagun for post-harvest processing, which provides farmers with technological solutions and training in handling and processing crops to maintain quality and extend shelf life¹⁶. There is also a push for the creation of pre-cooling infrastructure, reefer vans for cold chain maintenance, and primary processing facilities at aggregation centres¹⁷.

Arunachal Pradesh currently has minimal fruit and vegetable processing activity, mainly due to insufficient horticultural output to support large-scale units. It is recommended to set up processing units at strategic locations with good access to both production areas and external markets. These units should align their product lines with demand in major consumption centres outside the state. Multi-product facilities are preferred to ensure higher processing volumes and year-round operations. Crops like apple, pineapple, tomato, and orange offer vast potential for diverse products such as chips, jams, juice powders, purees, and concentrates, which have strong market demand. Notably, orange, tomato, and pineapple are among the most consumed juices globally. The report¹⁸ by NEDFi found that there is an opportunity to develop the following fruit and vegetable processing and allied units in the state by the private sector/ Co-operatives/ Producer Companies etc. with support from the central/ state government:

Table 17. Opportunities for investment in Perishable and Non-Perishable produce in Arunachal Pradesh

Sl. No.	Units	Crops	Location	Investment (Rs. Crore)
1	Multi-product Processing Unit (Pulping, Juice & Concentrate – 5.4 Lakh bottles/Annum)	Orange, Pineapple, Kiwi, Apple etc.	Bhalukpong, Pasighat, Tezu	2.5 × 3 Nos.

¹⁵ Shrishailam, B, et al (2025) Agri-Entrepreneurship Opportunities in Arunachal Pradesh and Supporting Government Schemes. Available at <https://justagriculture.in/wp-content/uploads/2025/04/01.-Agri-Entrepreneurship-Opportunities-in-Arunachal-Pradesh-and-Supporting-Government-Schemes.pdf>

¹⁶ <https://arunachaltimes.in/index.php/2019/08/21/csir-unit-set-up-in-arunachal-for-post-harvest-processing/>

¹⁷ <https://www.indiatodayne.in/arunachal-pradesh/story/arunachal-showcases-organic-potential-eyes-global-markets-at-international-meet-1198669-2025-04-14>

¹⁸ NEDFi – “Study and Action Plan for Promoting Fruits and Vegetable Processing Industries in the state of Arunachal Pradesh”

2	Pickle Manufacturing Unit	Fruits & Vegetables	Ziro	0.48
3	RTS Beverage (Fruit-Based Juices) Aseptic Packing	Orange, Pineapple, Kiwi, Apple, etc.	Nahar Lagun	9.2
4	Canning of Fruits & Vegetables and RTS in PET Bottles	Fruits & Vegetables	Itanagar	3.15
5	Vegetable and Fruit Frozen Unit - Individual Quick Freezing (IQF)	Fruits & Vegetables	Itanagar	4.05
6	Vegetable and Fruit Dehydration Unit	Fruits & Vegetables	Likabali (West Siang)/Pasighat	3.72
7	Spice Grinding and Packing Unit	Ginger, Large Cardamom, Chilli	Tezu, Pasighat	1.07 × 2 Nos.
8	Vegetable and Fruit Jam & Jelly Unit	Fruits & Vegetables	Likabali	0.23
9	Instant Dried Pickle	Fruits & Vegetables	Ziro	0.25
10	Potato Processing – Chips	Potato	Bhalukpong/Pasighat	0.35
11	Fruit Wine	Fruits	Pasighat	1.56
12	Papaya Candy	Papaya	Changlang	0.02
13	Osmo-dried Fruits	Fruits	Pasighat	0.58

Source: NEDFi Report, 2017

Cooperative farming and community resource-sharing are gaining ground, enabling bulk storage, collective marketing, and reduced input costs¹⁹.

Table 18. Agri processing/Industrial Cooperative in Arunachal Pradesh

District	Number of Agri processing/Industrial Cooperative
Changlang	8
East Siang	7
Kurung Kumey	3
Lohit	1
Lower Dibang Valley	6
Lower Subansiri	24
Namsai	3
Tawang	2
Tirap	1
Upper Subansiri	1
West Kameng	3
Grand Total	59

Source: National Cooperative Database, GOI

To overcome such issues, farm proximate aggregation centres at identified clusters are proposed by NEDFi with facilities such as modern sorting, grading, packaging and other crop specific minimal processing facilities. The clusters (and locations for aggregation centres) have been identified based on the field visits and survey work in the NEDFi report. The tentative locations and the facilities for aggregation centres are given below:

¹⁹ <https://arunachaltimes.in/index.php/2025/02/23/organic-agriculture-in-arunachal-prospects-challenges-and-the-way-forward/>

Table 19. Proposed aggregation centres at identified clusters (NEDFi)

District	Aggregation Centres	Crops	Facilities
Tawang	Namstering	Apple, Kiwi	Sorting, Grading, Pre-cooling
West Kameng	Dirang, Bomdila, Rupa, Kalaktang, Nafra, Shergaon	Apple, Kiwi, Walnut, Vegetables	Sorting, Grading, Pre-cooling, Walnut Dryers
East Siang	Pasighat, Mebo, Boleng	Orange, Pineapple, Large Cardamom, Ginger	Sorting, Grading, Pre-cooling, Cardamom and Ginger Dryers
Lower Dibang Valley	Roing, Dambuk	Ginger	Sorting, Grading, Ginger Dryers
Lohit	Balek, Wakro, Tezu, Chowkham	Orange, Pineapple, Orange, Ginger, Potato, Vegetables	Sorting, Grading, Ginger Dryers
Changlang	Changlang	Ginger, Vegetables	Sorting, Grading
Lower Subansiri	Ziro, Hapoli	Kiwi, Pineapple, Large Cardamom	Sorting, Grading, Large Cardamom Dryers
West Siang	Rungong, Along, Jomlo, Passing	Pineapple, Orange, Large Cardamom	Sorting, Grading, Large Cardamom Dryers

Source: NEDFi Report, 2017

The land needed for each aggregation centre is relatively modest, usually under 500–600 square meters. The government, in coordination with local farmers and communities who stand to benefit from this infrastructure, must undertake the precise identification of land for setting up these centres.

7.3.1 Primary Processing and Storage Facilities – Perishable produce

- Fruit pulp units:** For high-altitude fruits like kiwi, pineapple, and orange, pulp extraction and packaging help extend shelf life and enable downstream industries (juices, flavoured yogurt, etc.).
- Vegetable dehydration units:** Seasonal vegetables like beans, leafy greens, and mushrooms can be processed through solar or electric dehydration, enabling year-round sale.
- Juice and jam processing plants:** Particularly relevant for citrus (orange) and temperate fruits (apple, pear), enabling smallholder value addition and local entrepreneurship.
- Controlled Atmosphere (CA) Cold Storage:** Essential for temperate fruits such as apples, which are cultivated in districts like Tawang, West Kameng, and Lower Subansiri. CA storage extends shelf life up to 6 months, allowing farmers to time their market access better.
- Quick freezing units:** Required for perishables like green peas (grown in higher elevations) and floriculture produce, where flash-freezing preserves texture and quality for distant markets.

Case Example: Ziro Valley Kiwi Processing Hub: Establishment of a kiwi fruit processing unit that includes fruit sorting, pulp extraction, packaging, and CA cold storage. This hub can support farmer producer organizations (FPOs) and promote export-quality produce.

7.3.2 Primary Processing and Storage Facilities – Non-Perishable produce

- Dal mills:** Necessary for lentils and rajma, enabling the removal of husks and polishing for market sale.

- ii. **Oil expellers:** Beneficial for processing indigenous oilseeds like sesame and mustard, reducing transport costs and retaining value within the state.
- iii. **Grain grading and cleaning units:** To ensure quality assurance and better market prices, particularly important in tribal and remote regions.
- iv. **Warehousing with fumigation:** Scientifically designed warehouses are essential for long-term storage, protecting grains from pests, moisture, and fungal damage. They should include aeration and fumigation systems.

Case Example: Lohit Oilseed Processing Cluster: A dedicated crop-based unit in the Lohit district for sesame and mustard, integrating oil expelling, packaging, and branding. The facility can serve as a regional hub for Northeast oilseed marketing.

Annexure 4

Transport Connectivity

Arunachal Pradesh faces formidable challenges in the movement of agricultural produce due to its difficult terrain, limited transport infrastructure, and geographical isolation. While road transport remains the primary mode of connectivity, the lack of multimodal transport and cold-chain infrastructure significantly constrains agricultural value chains.

7.4.1 Road Transport

Road transport is the backbone of agricultural logistics in Arunachal Pradesh. The state is connected to national markets primarily through a network of highways that pass via Assam. Due to intra-state connectivity issues, goods often traverse through Assam before reaching other parts of Arunachal Pradesh. The Trans-Arunachal Highway, spanning 1,559 km, plays a crucial role in this network, linking the northwestern town of Tawang to Kanubari in the southeast. Despite its strategic importance, frequent landslides and monsoon-related disruptions render large parts of the state temporarily inaccessible, especially affecting perishable produce. The major routes and the districts they serve are outlined below:

Table 20. Major routes and the districts for Agriculture produce

S. No.	Road/Route	Districts Catered
1	Bomdila-Bhalukpong-Tezpur Road (NH-229)	West Kameng, Tawang, East Kameng
2	Ziro-Yupia-Tezpur Road (NH-229)	Lower Subansiri, Upper Subansiri, Papum Pare, Kurung Kumey
3	Yingkiong-Along-Pasighat Road (NH-229/ NH-52)	East Siang, West Siang, Upper Siang
4	Roing-Tezu-Wakro-Namsai (NH-52)	Lower Dibang Valley, Dibang Valley, Lohit, Changlang

Source: NEDFi Report, 2017

Key agricultural regions and travel estimates are given below:

Table 21. Key agricultural regions and travel estimates

Origin in Arunachal Pradesh	Main Route (via)	Approx. Distance (km)	Est. Time (hours)
Ziro (Lower Subansiri)	N. Lakhimpur, Tezpur	~450	11–13
Along/Aalo (West Siang)	Pasighat, Dibrugarh	~600–650	14–16
Pasighat (East Siang)	Dibrugarh, Jorhat	~550–600	13–15
Tawang (Namstering)	Bomdila, Tezpur	~520–550	15–18
Changlang (Changlang)	Tinsukia, Dibrugarh	~600–650	14–16

Source: The Infravision Foundation

Road transport forms the core of Arunachal Pradesh's agricultural logistics, with most connectivity to national markets routed through Assam due to limited intra-state road networks. The 1,559 km Trans-Arunachal Highway is vital for linking key districts, but frequent monsoon disruptions and landslides severely affect access—particularly for perishable produce. Major routes such as NH-229 and NH-52 serve critical agricultural regions, with average travel times to external markets ranging from 11 to 18 hours.

7.4.2 Rail Transport

Rail connectivity is minimal in Arunachal Pradesh. The state's first railway line, connecting Naharlagun to Assam, was inaugurated in 2015. While this offers some access to the national railway network, most farming communities remain distant from railheads. As a result, rail plays a negligible role in crop transport. Instead, produce is typically trucked to railway hubs in Assam such as North Lakhimpur or Siliguri for onward shipment. Railways offer a cheaper alternative for bulk transport; its expansion can result in a significant reduction in transport of agricultural produce.

7.4.3 Air Transport

Air transport is emerging as a potential enabler for high-value or perishable commodities. The inauguration of the Donyi Polo Airport near Itanagar in 2022 marks a turning point in improving connectivity to metro cities. However, air cargo services remain limited and there is no dedicated cold cargo terminal in Arunachal Pradesh. Exporters often depend on air freight services from Assam for shipments, limiting the state's direct access to premium markets.

7.4.4 Inland Waterways

Although Arunachal Pradesh has several navigable rivers like the Lohit, Subansiri, and Noa Dihing, their role in agricultural transport is currently limited. However, the broader Northeastern Region benefits from waterway infrastructure, particularly on the Brahmaputra (NW-2) and Barak River (NW-16), with future potential to integrate Arunachal into the inland water transport network.

7.4.5 Summary on Transport and Market Linkage Infrastructure

Arunachal Pradesh's agricultural logistics face critical constraints due to hilly terrain, limited multimodal infrastructure, and geographical isolation. Road transport remains the primary mode of connectivity, with major produce routes such as NH-229 and NH-52 linking key districts to Assam before accessing national markets. However, intra-state travel is often disrupted by landslides and monsoon-related events, especially impacting the movement of perishable produce. To address this, both perishable and non-perishable supply chains require aggregation hubs, rural roads with all-weather surfaces, and integrated digital logistics platforms linked to Farmer Producer Organizations (FPOs) for scale efficiency.

Perishables like vegetables, fruits, and flowers need an added layer of infrastructure including reefer trucks, packhouses, and cold storage points at transit hubs such as railheads or airports. For instance, a cold chain corridor from Dirang to Itanagar or Guwahati can reduce spoilage and improve market access. Rail transport remains underdeveloped, with only Naharlagun linked to Assam, pushing most goods to rely on trucks for access to distant railheads. Similarly, air cargo potential remains untapped due to the lack of dedicated cold terminals despite the opening of Donyi Polo Airport. Inland waterways offer future promise but are currently underutilized in Arunachal's agricultural value chains.

Annexure 5

State-wise description of the main fruits, vegetables and spices produced in the NER

State	Fruits	Vegetables	Spices
Arunachal Pradesh	Apple, Banana, Guava, Kiwi, Orange, Pear, Pineapple, Walnut	Bean, Bitter gourd, Brinjal, Cucumber, Muskmelon, Potato, Pumpkin, Radish, Sweet Potato, Tomato	Ginger, Large Cardamom
Assam	Banana, Guava, Jack fruit, Lemon, Litchi, Mandarin Orange, Mango, Papaya, Pineapple, Sapota	Bean, Brinjal, Broccoli, Cabbage, Capsicum, Carrot, Cauliflower, Colocasia, Cucumber, Gourds (different types), Knol-khol, Muskmelon, Pea, Potato, Pumpkin, Radish, Spinach, Sweet potato, Tapioca, Tomato, Yam	Black cumin, Black pepper, Chillies, Coriander, Cumin, Fennel, Fenugreek, Garlic, Ginger, Mint, Onion, Turmeric
Manipur	Banana, Lemon, Orange, Passion fruit, Peach, Pear, Pineapple, Plum	Bean, Cabbage, Cauliflower, Cucumber, Muskmelon, Pea, Potato, Tomato	Chillies, Garlic, Ginger, Hatkora, Onion, Turmeric
Meghalaya	Banana, Guava, Lemon, Mandarin Orange, Pear, Pineapple, Plum	Broccoli, Cabbage, Capsicum, Carrot, Cauliflower, Colocasia, Potato, Radish, Squash, Sweet potato, Tapioca, Tomato, Turnip	Bay leaf, Black pepper, Chillies, Ginger, Turmeric
Mizoram	Banana, Hatkora, Orange, Papaya, Passion fruit, Pineapple	Bean, Brinjal, Broccoli, Cabbage, Capsicum, Carrot, Cauliflower, Chow-chow, Cucumber, different types of Gourds, Knol-khol, Ladies finger, Muskmelon, Pea, Potato, Pumpkin, Radish, Tomato	Chillies, Ginger, Turmeric
Nagaland	Banana, Guava, Jack fruit, Lemon, Litchi, Mandarin Orange, Mango, Papaya, Pineapple, Plum, Pomegranate	Brinjal, Cabbage, Carrot, Cauliflower, Chow-chow, Colocasia, Ladies finger, Pea, Potato, Radish, Sweet Potato, Tomato	Black pepper, Cardamom, Chillies, Garlic, Ginger, Turmeric
Sikkim	Banana, Kiwi, Mandarin Orange, Papaya, Passion fruit, Pear	Bean, Broccoli, Cabbage, Cauliflower, Cucumber, Pea, Radish, Tomato, Turnip	Cherry pepper, Ginger, Large cardamom, Turmeric
Tripura	Banana, Ber, Guava, Jack fruit, Lemon, Litchi, Mango, Musambi, Orange, Papaya, Pineapple, Sapota	Bean, Cabbage, Capsicum, Carrot, Cauliflower, Cucumber, different types of Gourds, Parmal, Potato, Pumpkin, Radish, Sweet potato, Potato, Tapioca, Tomato	Black Pepper, Betel vine, Chillies, Ginger, Onion

Source: *Journal of Horticulture Science* Vol. 17(2): 530-542, 2022

Annexure 6

State-wise distribution of numbers of operational holdings and area operated in the NER – all social groups

(Number: 000', Area: 000' ha)

Marginal: below 1ha, Small: 1 to 2ha, Semi-medium 2 to 4 ha, Medium: 4 to 10 ha, Large: above 10 ha

States/ Union Territories	Agriculture Census	Marginal		Small		Semi-medium		Medium		Large		All Holdings	
		No.	Area	No.	Area	No.	Area	No.	Area	No.	Area	No.	Area
Arunachal Pradesh	2010-11	21	12	19	26	34	94	28	155	7	97	109	384
	2015-16	27	14	24	31	29	77	26	153	7	104	113	380
Assam	2010-11	1831	775	497	687	304	818	85	437	4	282	2720	2999
	2015-16	1868	785	495	696	295	806	79	410	4	279	2742	2976
Manipur	2010-11	77	40	49	63	22	55	3	13	Neg	Neg	151	172
	2015-16	77	40	49	63	22	55	3	13	Neg	Neg	150	172
Meghalaya	2010-11	103	46	58	77	41	113	8	47	Neg	4	210	287
	2015-16	123	56	60	80	40	109	9	51	Neg	4	232	300
Mizoram	2010-11	50	30	30	38	10	24	2	9	Neg	4	92	105
	2015-16	45	27	27	35	14	32	3	15	Neg	4	90	112
Nagaland	2010-11	6	3	20	23	48	125	78	481	25	442	178	1074
	2015-16	8	4	29	36	63	168	74	435	23	355	197	998
Sikkim	2010-11	40	15	17	20	11	27	6	32	1	12	75	107
	2015-16	44	18	13	18	8	20	2	14	Neg	6	68	76
Tripura	2010-11	499	140	55	76	22	54	3	14	Neg	1	578	285
	2015-16	504	149	48	70	19	49	2	13	Neg	1	573	282

Source: Horticulture Statistics at a Glance, 2021, GOI

Annexure 7

State-wise Area and Production of Horticulture Crops from the year 2011-12 to 2020-21

Area in '000 Ha, Production in '000 MT

Years States/UT's	2011-12		2012-13		2013-14		2014-15		2015-16	
	Area	Prod.	Area	Prod.	Area	Prod.	Area	Prod.	Area	Prod.
ARUNCHAL PRADESH	107.87	563.14	103.72	523.26	106.83	532.14	107.04	547.72	83.23	417.41
ASSAM	596.32	5427.05	626.02	5971.45	624.31	5546.26	639.76	7115.21	671.68	6481.49
MANIPUR	80.77	630.31	84.13	684.55	91.37	812.66	97.13	814.17	97.02	808.95
MEGHALAYA	111.23	789.08	113.58	823.94	122.07	978.99	124.1	1027.06	127.89	1014.07
MIZORAM	106.93	626.3	120.26	761.16	130.03	835.75	139.68	882.2	137.44	586.97
NAGALAND	77.62	611.07	74.54	533.37	90.56	954.24	91.06	954.33	96.98	995.05
SIKKIM	63.01	230.48	67	243.11	74.43	232.26	63.49	209.86	67.48	212.1
TRIPURA	110.09	1256.23	126.18	1503.49	136.57	1617.11	141.95	1680.22	144.07	1698.56

Years States/UT's	2016-17		2017-18		2018-19		2019-20		2020-21	
	Area	Prod.	Area	Prod.	Area	Prod.	Area	Prod.	Area	Prod.
ARUNCHAL PRADESH	62.46	208.7	58.49	177.62	63.16	173.17	66.68	172.94	67.53	176.8
ASSAM	664.42	5914	659.34	6000.9	656.84	6242	667.61	6432.6	650.33	6467.5
MANIPUR	121.41	872.37	105.81	828.88	105.11	840.47	91.37	947.71	82.91	877.43
MEGHALAYA	131.01	1075.93	126.11	946.70	129.72	1027.03	142.46	1021.00	140.70	1006.60
MIZORAM	137.42	625.02	139.98	623.41	138.36	660.82	146.34	647.27	156.68	705.98
NAGALAND	103.57	1069.92	97.79	1016.06	88.35	847.17	88.57	846.60	89.84	829.31
SIKKIM	76.58	299.80	90.27	367.57	101.20	410.10	101.20	410.42	79.80	278.99
TRIPURA	123.77	1431.50	121.10	1421.61	122.92	1430.47	124.16	1444.48	126.86	1485.72

Source: Horticulture Statistics at a Glance, 2021, GOI

